Chapter 11
MOSFET: Additional Concepts

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11.0 Preview

1. Describe and analyze subthreshold conduction, which is the phenomenon whereby current is induced in the channel before the defined threshold voltage is reached.
2. Analyze channel length modulation, which is a characteristic of short-channel lengths and leads to a finite output resistance.
3. Consider the effects of a decrease in carrier mobility due to increasing gate voltage.
4. Analyze the effects of carrier saturation velocity. Carriers can easily reach their saturation velocity in short-channel devices.
5. Discuss MOSFET scaling, which describes how various parameters must be changed as device size is decreased.
6. Consider the deviations in threshold voltage due to small geometry devices, including short channel length devices and small channel width devices
7. Describe and analyze various voltage breakdown mechanisms in MOSFETs.
8. Describe and analyze the technique of threshold voltage adjustment by ion implantation.
9. Consider the introduction of trapped oxide charges by ionizing radiation and hot electron effects

11.1 Nonideal Effects

11.1.1 Subthreshold Conduction

11.1.2 Channel Length Modulation

11.1.3 Mobility Variation

11.1.4 Velocity Saturation

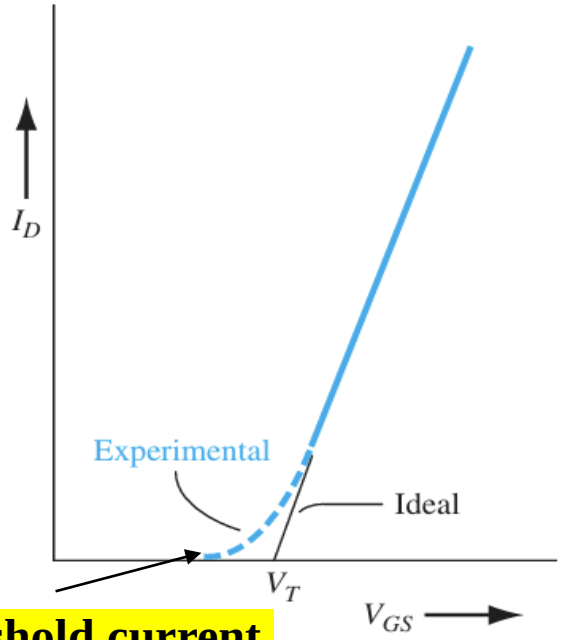
11.1.5 Ballistic Transport

11.1.1 Subthreshold Conduction

Subthreshold current: MOSFET drain $\boldsymbol{I_D}$ is not zero when $V_{GS} \leq V_T$ for experimental or real devices.

	$V_{GS} \leq V_T$
Ideal MOSFET	$I_D=0$
Real MOSFET	$I_D>0$

Why?



Subthreshold current

Figure 11.1 | Comparison of ideal and experimental plots of $\sqrt{I_D}$ versus V_{GS}.

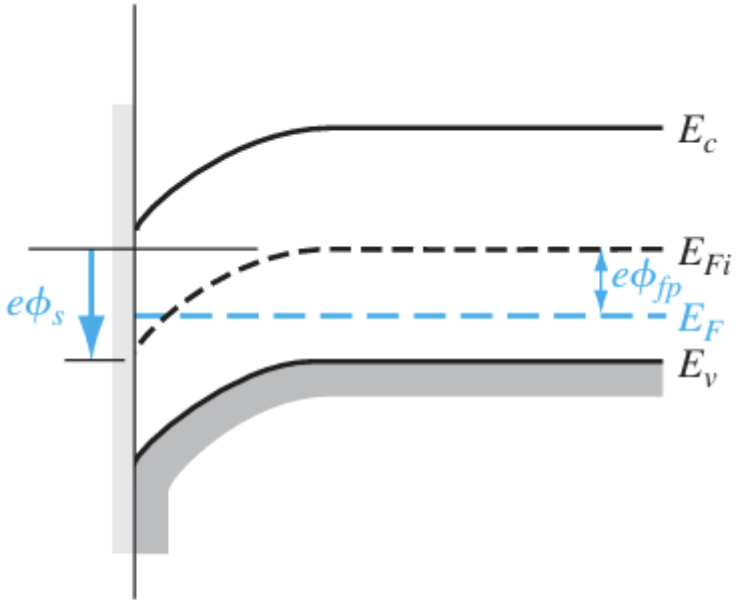


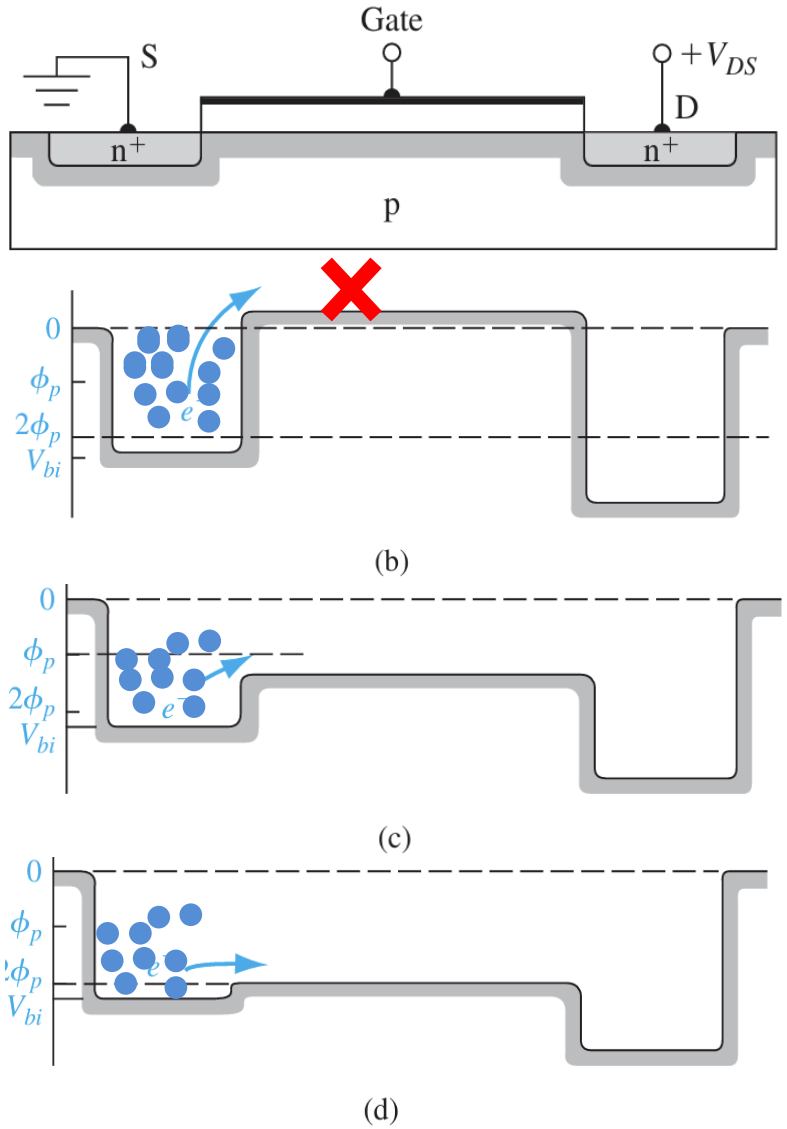
Figure 11.2 | Energy-band diagram when $\phi_{fp} < \phi_s < 2\phi_{fp}$.

The biased($\varphi_s \leq 2\varphi_{fp}$) P-sub MOS Energy band shows: the Fermi level is closer to the conduction band than the valence band at surface, similar to a lightly doped n-type material.

Due to the N source and drain contacts through this weakly inverted channel in P-sub MOSFET, little $\boldsymbol{I_D}$ was catch and this condition of ($\varphi_{fp} \leq \varphi_s \leq 2\varphi_{fp}$) was named ***weak inversion*.**

11.1.1 Subthreshold Conduction

The surface potential along the length of the channel at **accumulation, weak inversion,** and **threshold** for NMOSFET with different V_{DS}.



The surface potential diagram of channel when the NMOSFET was biased small V_{DS}.

Accumulation: **high** potential barrier→ electrons cannot jump→ **no** surface current

Weak inversion: **low** potential barrier→ individual electrons cross the barrier → **little** subthreshold current I_D

Threshold: **lower or lowest** potential barrier→ most electrons cross the barrier→ **large** channel current I_D

11.1.1 Subthreshold Conduction

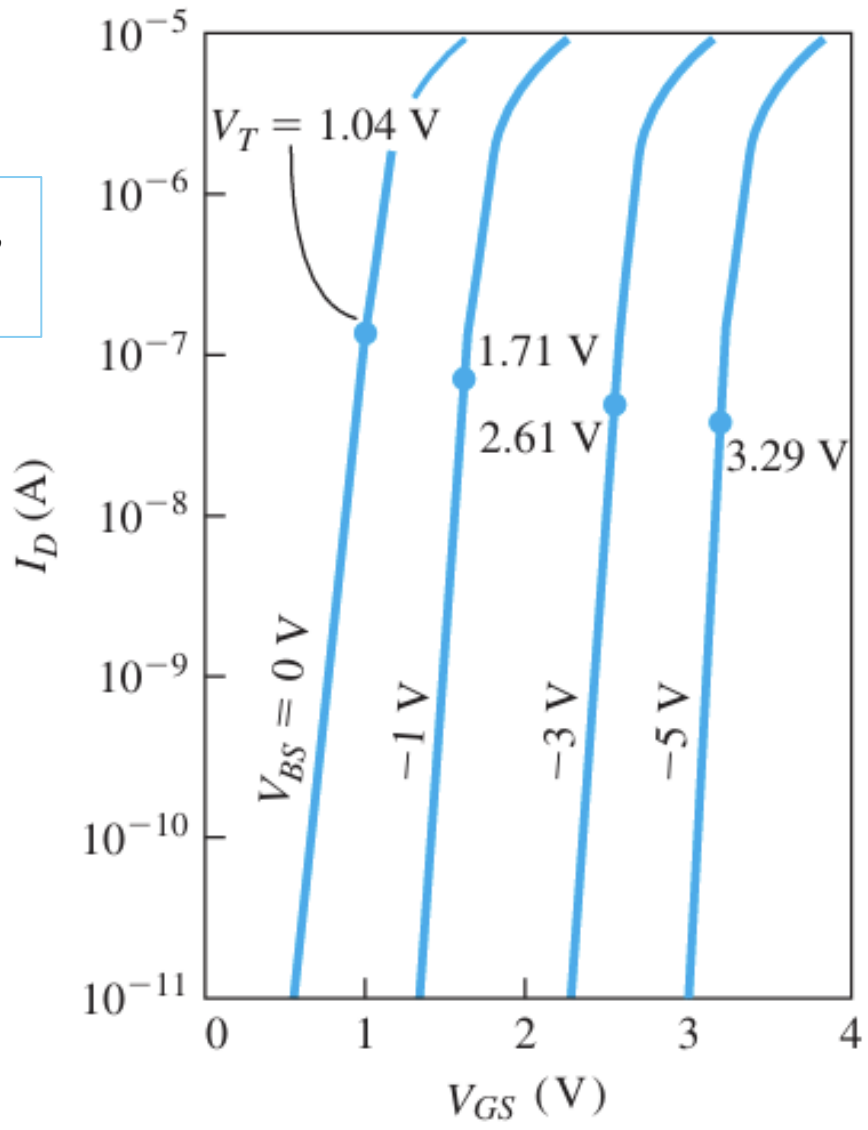
Subthreshold current voltage curves for different V_{BS}

$$I_D(\text{sub}) \propto \left[\exp\left(\frac{eV_{GS}}{kT}\right)\right] \cdot \left[1 - \exp\left(\frac{-eV_{DS}}{kT}\right)\right]$$

If V_{DS} is larger than a few (kT/e) volts (example: $T = 300\text{K}$, $V_{DS} > \frac{4e}{kT} = 0.104V$), then the subthreshold current I_D is independent of V_{DS}.

$I_{D(sub)}$ was exponentially related to the V_{GS}, which means that 60 mV change of V_{GS} will result in an order magnitude change of the $I_{D(sub)}$.

The slope of the I_{Dsub}-V_{DS} curve is a function of the doping concentration and the density of states at the interface. The density of states at the oxide-semiconductor interface can be determined experimentally by measuring the slope of the curve.



11.1.2 Channel Length Modulation

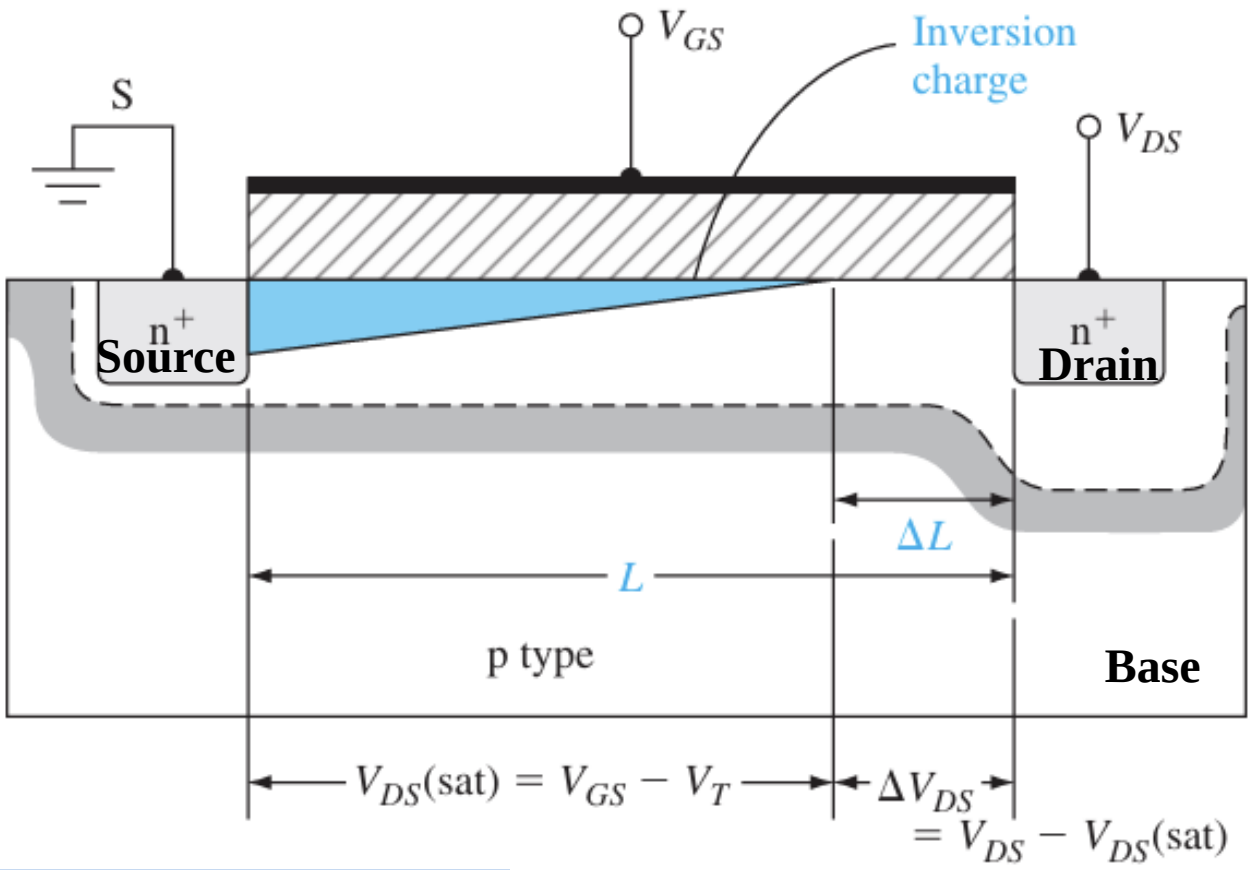
Channel length modulation effect: When MOSFET at the **saturation region**, the depletion region width was modulated by V_{DS}, and at the drain terminal extends laterally into the channel, reducing the effective channel length, result in the effective channel length was modulated by V_{DS}.

For **PN junction**, V_{bias}= 0V

the depletion p-region width: $x_p = \sqrt{\frac{2\varepsilon_S \phi_{fp}}{eN_a}}$

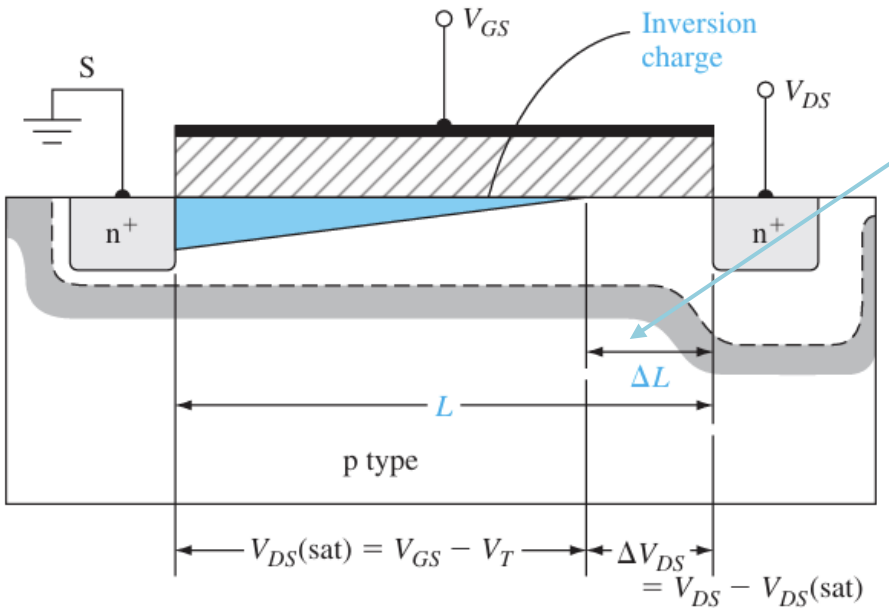
For one-sided N^+P junction (**Drain–Base DB**), all reverse voltage was across the low-doped p region.

the space charge width: $x_p = \sqrt{\frac{2\varepsilon_S}{eN_a}(\phi_{fp} + V_{DS})}$



Saturation region: $V_{DS} \uparrow \Rightarrow \Delta V_{DS} \uparrow \Rightarrow \Delta L \uparrow \Rightarrow L' = L - \Delta L \downarrow \Rightarrow I_D \uparrow \left(\propto \frac{1}{L'}\right)$

11.1.2 Channel Length Modulation



$$\Delta V_{DS} = V_{DS} - V_{DS(sat)}$$

the expand space charge region of the DB pn junction

channel cutoff V_{DS}	the space charge region width of the DB pn junction	$V_{DB} =$
$= V_{DS(sat)}$	$x_{p1} = \sqrt{\frac{2\varepsilon_S}{eN_a}(\phi_{fp} + V_{DS(sat)})}$	$\phi_{fp} + V_{DS(sat)}$
$> V_{DS(sat)}$	$x_{p2} = \sqrt{\frac{2\varepsilon_S}{eN_a}(\phi_{fp} + V_{DS(sat)} + \Delta V_{DS})}$	$\phi_{fp} + V_{DS(sat)} + \Delta V_{DS}$

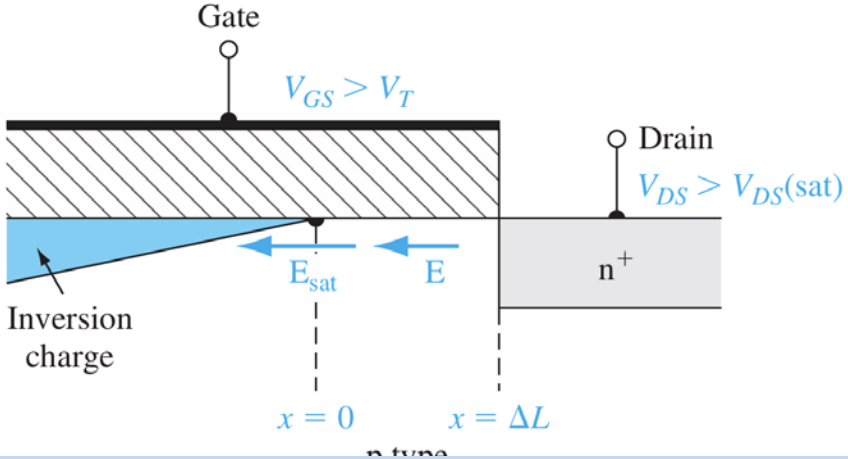
the space charge region width : $\Delta L = x_{p2} - x_{p1} = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\varphi_{fp} + V_{DS(sat)} + \Delta V_{DS}} - \sqrt{\varphi_{fp} + V_{DS(sat)}}\right]$

11.1.2 Channel Length Modulation

In cutoff region:

the electric field one-dimensional Poisson's equation: $\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon_s}$ (1)

Charge density :$\rho(x) = -eN_a$ (constant for uniformly doped base)



the electric field: $E(x) = \int_0^x \frac{\rho}{\varepsilon_s} dx - E_{sat} = -\frac{eN_a x}{\varepsilon_s} - E_{sat}$; ; (Integrating (1) with the boundary conditions $E|_{x=0} = E_{sat}$);

the potential: $\varphi(x) = -\int_0^x E(x)dx = \frac{eN_a x^2}{2\varepsilon_s} + E_{sat}x + V_{DS(sat)}$; (the boundary conditions $\varphi|_{x=0} = V_{DS(sat)}$)

$$\Delta V_{DS} = \int_0^{\Delta L} E(x)\, dx = \varphi(\Delta L) - \varphi(0) = \frac{eN_a \Delta L^2}{2\varepsilon_s} + E_{sat}\Delta L$$

$$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\varphi_{sat} + (V_{DS} - V_{DS(sat)})} - \sqrt{\varphi_{sat}}\right];$$

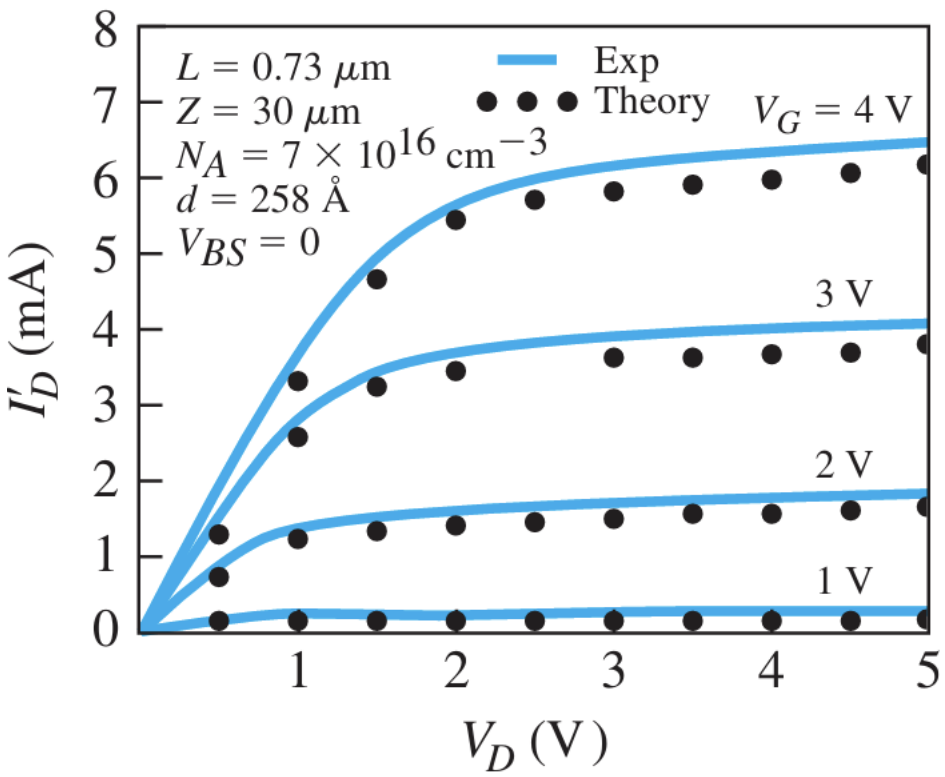
$$\varphi_{sat} = \frac{2\varepsilon_s}{eN_a}\left(\frac{E_{sat}}{2}\right)^2$$

$$I_{D(sat)} = \frac{W\mu_n C_{ox}}{2L}(V_{GS} - V_T)^2$$

Actual drain to source $I_D' = \left(\frac{L}{L-\Delta L}\right) I_{D(sat)} > I_{D(sat)}$

In general, E_{sat} is in the range $10^4 < E_{sat} < 2*10^5$ V/cm.

11.1.2 Channel Length Modulation



Actual drain to source $I_D{}' = \left(\frac{L}{L-\Delta L}\right) I_{D(sat)} > I_{D(sat)}$

1. **Actual $I'_D > I_{D(sat)}$**
2. **the value of I'_D increased with VDS in saturation region**

$$\Delta L = \sqrt{\frac{2\varepsilon_s}{eN_a}}\left[\sqrt{\varphi_{sat} + \left(V_{DS} - V_{DS(sat)}\right)} - \sqrt{\varphi_{sat}}\right];$$

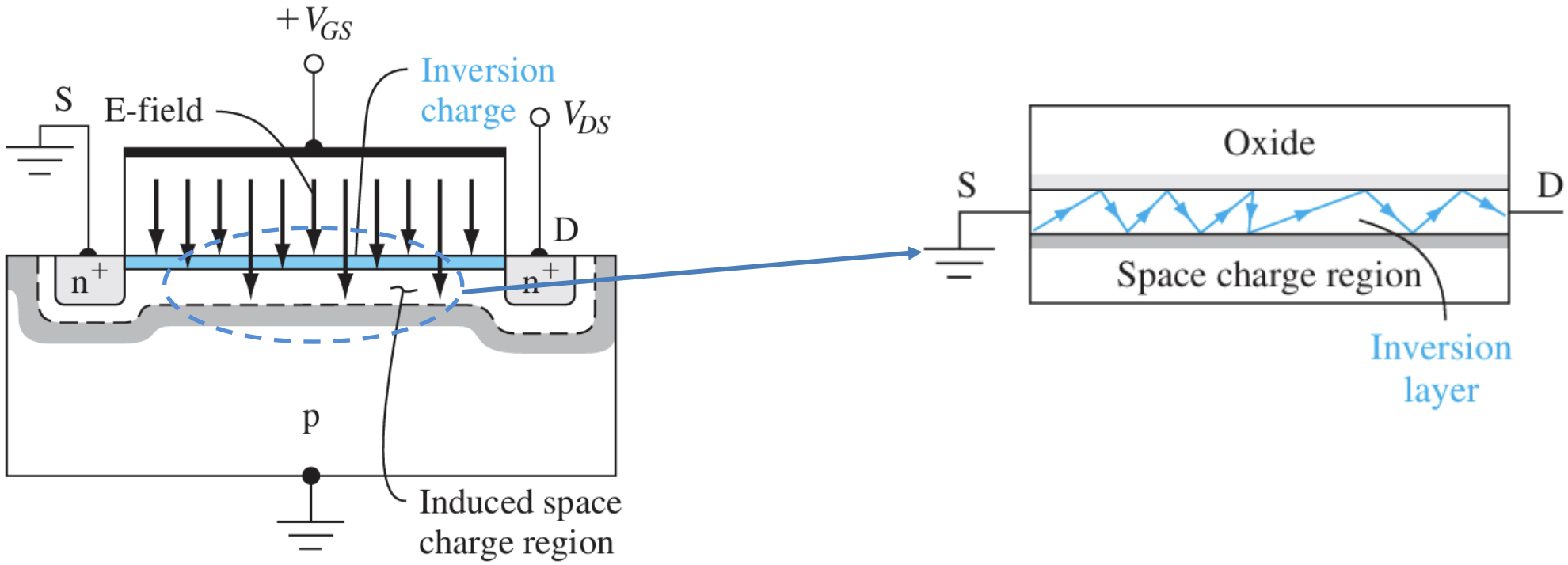
Smaller doping concentration Na in base → greater absolute value of ΔL

Smaller channel length L → the greater relative value of ΔL

more severe the channel length modulation effect

11.1.3 Mobility Variation

surface scattering effect reduces mobility



V_{GS}↑ → vertical electric field ↑ →the carriers transport near the channel interface → Surface scattering↑ → Surface mobility μ_s↓→ $\mu=f(V_{GS})$

11.1.3 Mobility Variation

surface scattering effect reduces mobility

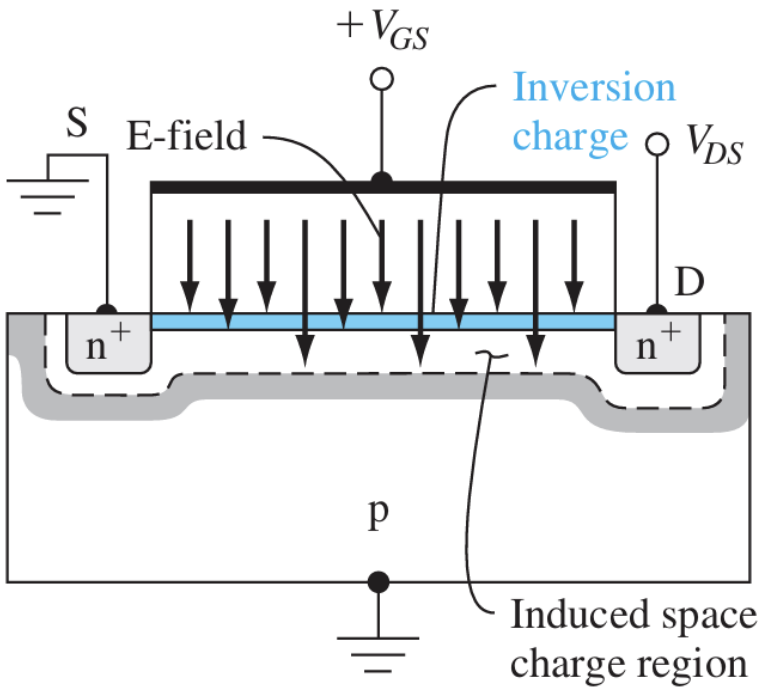
The effective mobility: $\frac{1}{\mu_{eff}} = \frac{1}{\mu_0} + \frac{1}{\mu_s}$

Volume mobility (typical 600cm²/Vs, NMOS)

surface mobility

$$\mu_{eff} = \frac{\mu_0}{1 + \frac{\mu_0}{\mu_s}} = \frac{\mu_0}{1 + \theta(V_{GS} - V_T - V(x))} = f(V_{GS}, x)$$

typical:0.03



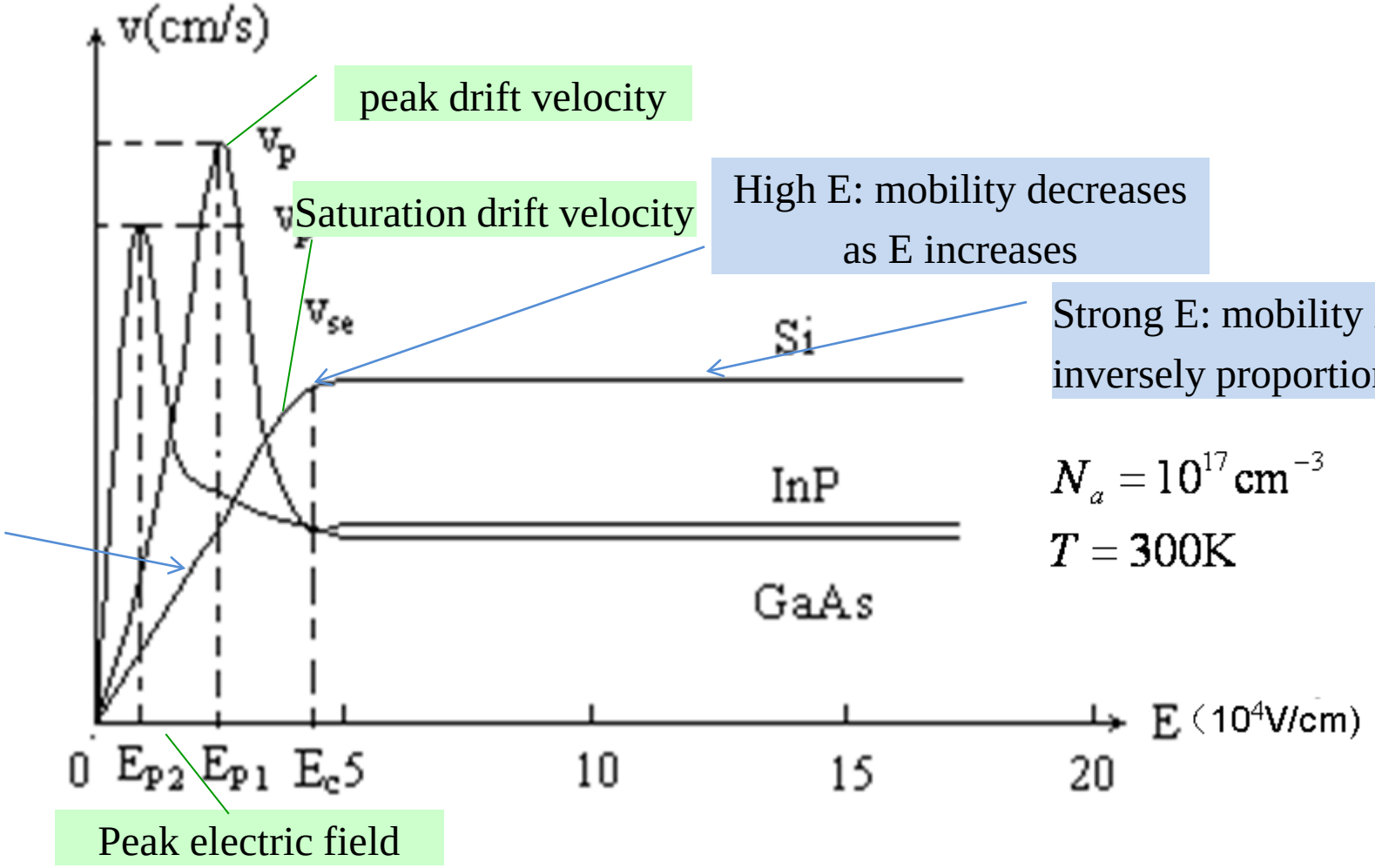
$$I_D = \frac{WC_{ox}}{L}\int_0^{V_{DS}} \frac{\mu_0}{1 + \theta[V_{GS} - V_{Tn} - V(x)]}[V_{GS} - V_{Tn} - V(x)]dV_x = \frac{\beta}{\theta}\left[V_{DS} + \frac{1}{\theta}\ln\left(\frac{1 + \theta(V_{GS} - V_{Tn} - V_{DS})}{1 + \theta(V_{GS} - V_{Tn})}\right)\right]$$

V_{GS} -V_{Tn} ↑ → I_D ↑ slowly

11.1.3 Mobility Variation

Drift velocity VS electric field for **Si**

$$v = \mu E$$
$$\mu = \frac{dv}{dE}$$



High E: mobility decreases as E increases

Strong E: mobility is inversely proportional to E

$N_a = 10^{17}\,\mathrm{cm}^{-3}$

$T = 300\mathrm{K}$

Low E: mobility independent with E

$$E \propto \frac{V_{DS}}{L} \begin{cases} \text{easily saturate in short channels;} \quad (V_{DS} = 5\mathrm{V}, L = 1\mu\mathrm{m},\ \mathrm{E} = 5 \times 10^4 \mathrm{V/cm}) \\ \text{easily saturate at high voltage} \end{cases}$$

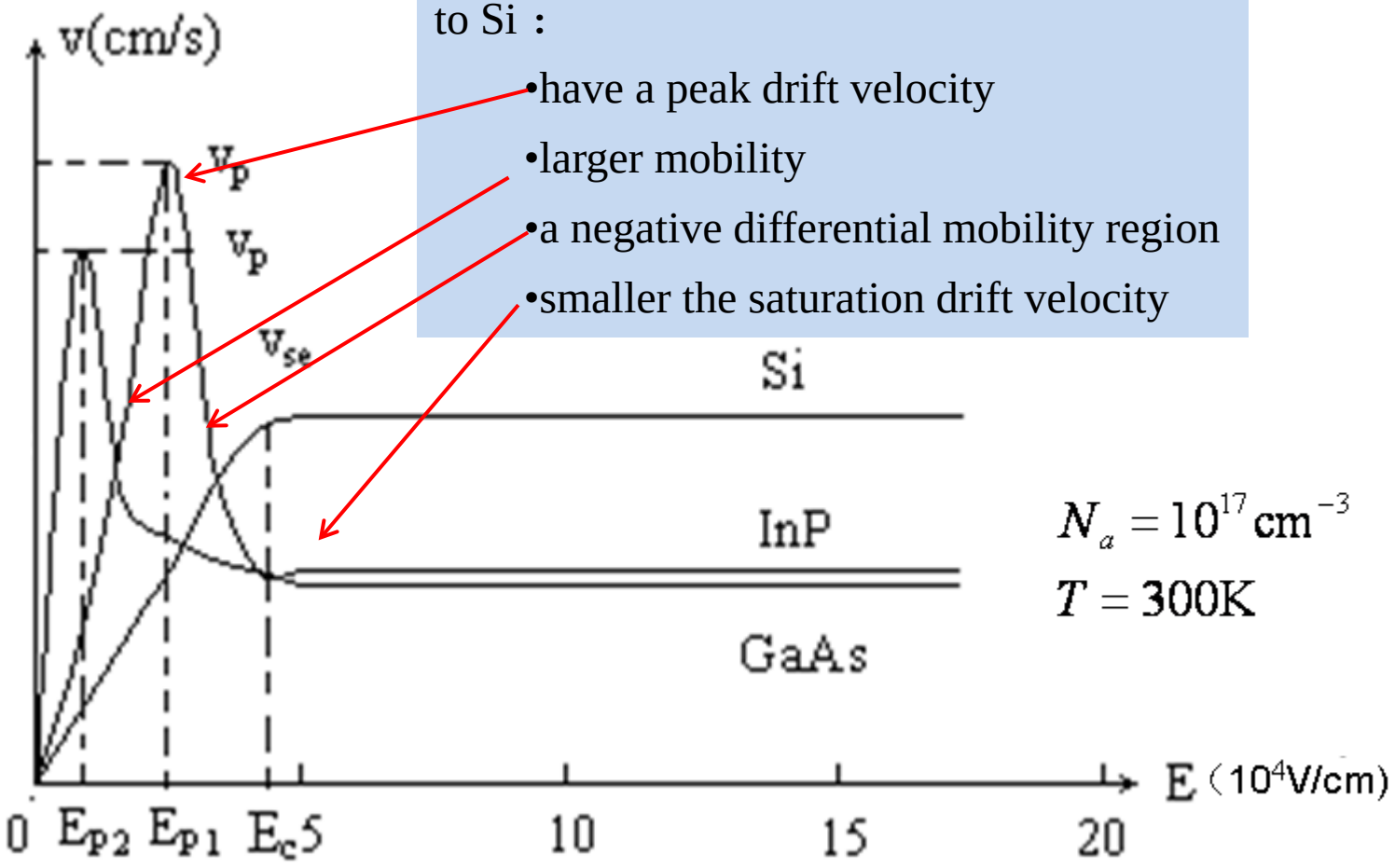
11.1.3 Mobility Variation

Drift velocity VS electric field for **GaAs、InP**

Si: $v_{se} = 1 \times 10^7$cm/s, $E_C = 5 \times 10^4$cm/V
GaAs: $v_p = 2 \times 10^7$cm/s, $E_p = 3.5 \times 10^3$cm/V
$v_{se} = (6\sim8) \times 10^6$cm/s
InP: $v_p = 3 \times 10^7$cm/s, $E_p = 1.1 \times 10^3$cm/V
$v_{se} = (6\sim8) \times 10^6$cm/s

Characteristics of GaAs and InP compared to Si：

- have a peak drift velocity
- larger mobility
- a negative differential mobility region
- smaller the saturation drift velocity



11.1.4 Velocity Saturation

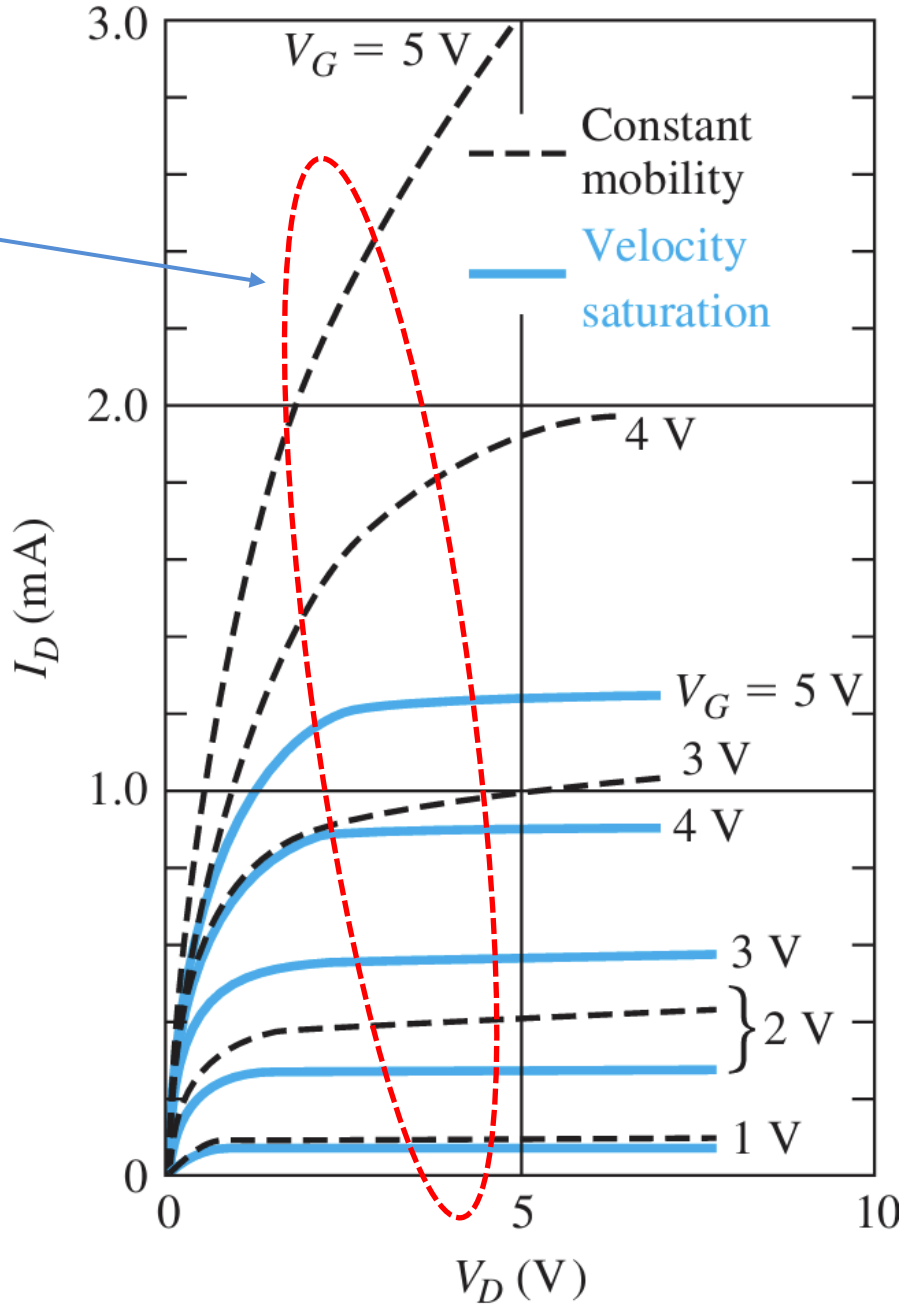
I_D decrease

Saturation in advance: $V_{DS(sat)} < V_{GS} - V_T$

$I_{D(sat)}$ is linearly related to V_{GS}: $I_{D(sat)} = WC_{OX}(V_{GS} - V_T)v_{sat}$

$g_{m(sat)}$ is independent of V_{DS} and L: $g_{m(sat)} = \frac{\partial I_{D(sat)}}{\partial V_{GS}} = WC_{OX}v_{sat}$

fT *is independent* of V_{GS} $f_T = \frac{g_m}{2\pi C_G} = \frac{WC_{OX}v_{sat}}{2\pi(C_G WL)} = \frac{v_{sat}}{2\pi L}$



11.1.5 Ballistic Transport

As the MOSFET channel length L is reduced, the mean distance between collisions l may become comparable to L OR less than L, then a large fraction of carriers could travel from the source to the drain without experiencing a scattering event. This motion of carriers is called **Ballistic Transport**.

- Non-ballistic transport MOSFET
  - L>0.1μm, longer than the mean distance between collisions
  - Carriers were scattered many times during transport from source to drain
  - The carrier was characterized by the average drift velocity
- Ballistic transport MOSFET
  - L<0.1μm, less than the mean distance between collisions
  - a large fraction of carriers could travel from the source to the drain without experiencing a scattering event
  - High-speed devices, Nano devices

11.2 MOSFET SCALING

Why Shrink the MOSFET?

11.2.1 Constant-Field Scaling
11.2.2 Threshold Voltage—First Approximation
11.2.3 Generalized Scaling

Higher integration: less chip area required for the same function;

More functions: more functions can be achieved in the same area;

Lower cost: the cost of a single tube is reduced;

Stronger performance: faster speed, lower unit power consumption;

•If the size was shrink by 30%, then

- Gate delay is reduced by 30% and operating frequency is increased by 43%;
- Doubling the number of transistors per unit area;
- The energy required for each switching is reduced by 65%, and the power consumption is reduced by 50%;

11.2 MOSFET SCALING

How to shrink?

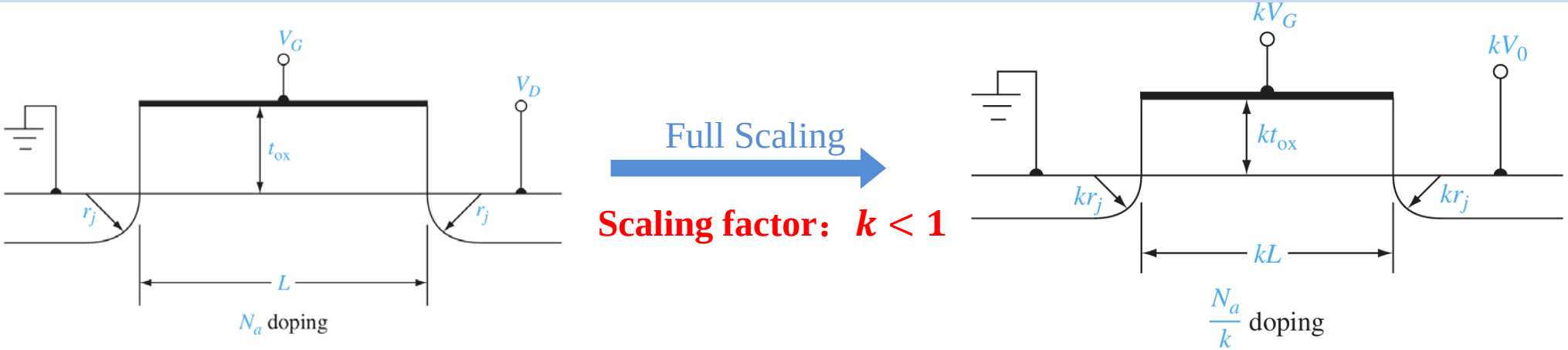
- **Full Scaling**
  - The voltage is reduced in the same proportion as the size ;
  - The electric field strength remains constant;
  - Ideal, but difficult to achieve;

- **Fixed Voltage Scaling**
  - The size shrinks proportionally, and the voltage is unchanged;
  - The electric field strength increases with decreasing size, and the strong field effect increases

- **General Scaling**
  - The size and the electric field are reduced by different proportional factors;
  - Practical practices so far;

11.2.1 Constant-Field Scaling

The principle of **constant-field scaling** is that device **dimensions and voltages be scaled** such that electric fields **E** (both horizontal and vertical) remain essentially **constant**.



$$W' = kW, L' = kL \xrightarrow{unchange\ \mathrm{E_x}} V_{DS}' = kV_{DS} \xrightarrow{\text{unchanged } I-V} V_{GS}' = kV_{GS}, V_T' = kV_T$$

$$\xrightarrow{unchange \mathrm{E_y}} t_{ox}' = kt_{ox},\ x_D' = kx_D \xrightarrow{x_D \propto \sqrt{\frac{V_{DS}}{N_a}}} N_a' = N_a/k$$

$$x_D = \sqrt{\frac{2\epsilon(V_{bi} + V_D)}{eN_a}}$$

The drain current per channel width, for the transistor biased in the saturation region:

$$I_D = \frac{W\mu_n C_{ox}}{2L}[2(V_{GS} - V_T)V_{DS} - V_{DS}^2]$$

$$\frac{I_D}{W} = \frac{\mu_n \epsilon_{ox}}{2t_{ox}L}(V_G - V_T)^2 \rightarrow \frac{\mu_n \epsilon_{ox}}{2(kt_{ox})(kL)}(kV_G - V_T)^2 \approx \text{constant}$$

11.2.1 Constant-Field Scaling

Result

channel $\frac{W}{L}' = \frac{W}{L}$

Channel area: $W'L' = k^2 WL$

On resistance $R'\left(\propto \frac{L}{W}\right) = R$

All gate Capacitance $C_{ox}'\left(\propto \frac{WL}{t_{ox}}\right) = kC_{ox}$

S to D current: $I_D'\left(\propto \frac{W}{L}V\right) = kI_D$

Circuit delay time $\tau'(\propto RC) = k\tau$

Power density: $p'\left(\propto \frac{P}{WL}\right) = p$

Power dissipation per device $P'(\propto IV) = k^2 I_D$

11.2.1 Constant-Field Scaling

Summary

Table 11.1 | Summary of constant-field device scaling

	Device and circuit parameters	Scaling factor ($k < 1$)
Scaled parameters	Device dimensions (L, t_{ox}, W, x_j)	k
	Doping concentration (N_a, N_d)	$1/k$
	Voltages	k
Effect on device parameters	Electric field	1
	Carrier velocity	1
	Depletion widths	k
	Capacitance ($C = \epsilon A/t$)	k
	Drift current	k
Effect on circuit parameters	Device density	$1/k^2$
	Power density	1
	Power dissipation per device ($P = IV$)	k^2
	Circuit delay time ($\approx CV/I$)	k
	Power−delay product ($P\tau$)	k^3

Source: Taur and Ning [23].

11.2.2 Threshold Voltage—First Approximation

For a uniformly doped base:

$$V_T = \frac{\sqrt{2e\varepsilon N_a(2\varphi_{fp})}}{C_{ox}} + V_{FB} + 2\varphi_{fp}$$ (the first part $\propto \sqrt{k}$, latter parts have no relation to k)

So V_T does not scale directly with the scaling factor k

11.2.3 Generalized Scaling

In constant-field scaling, V_{bias} scaled with the same scaling factor k; actual technology evolution, voltages have not been reduced with the same scaling factor

11.3 THRESHOLD VOLTAGE MODIFICATIONS

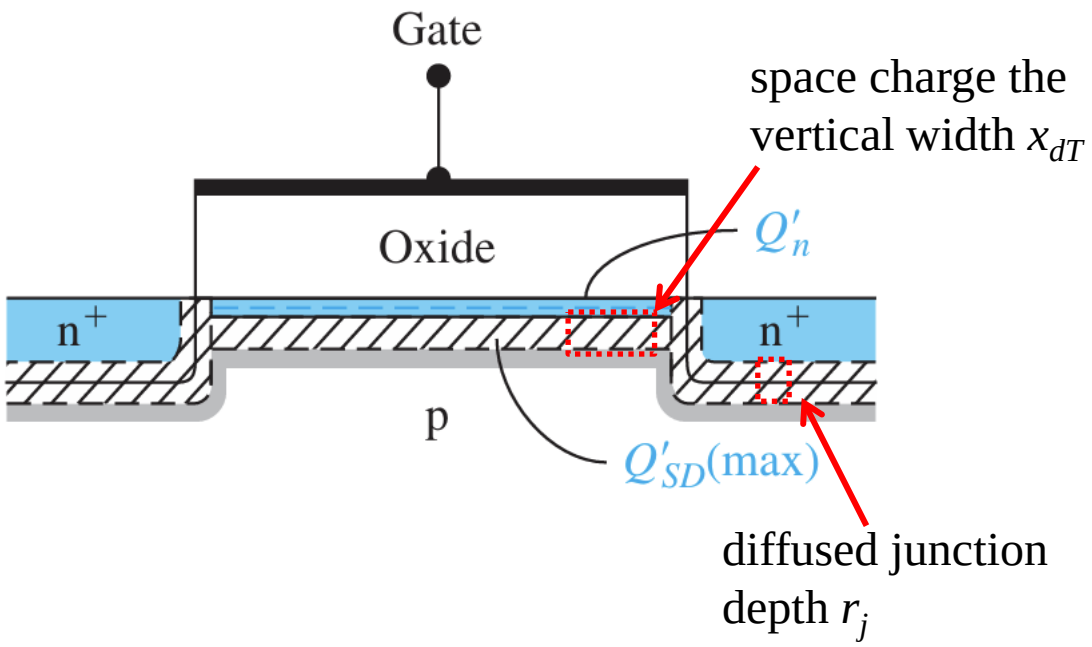
V_T varies with L or W

11.3.1 Short-Channel Effects
11.3.2 Narrow-Channel Effects

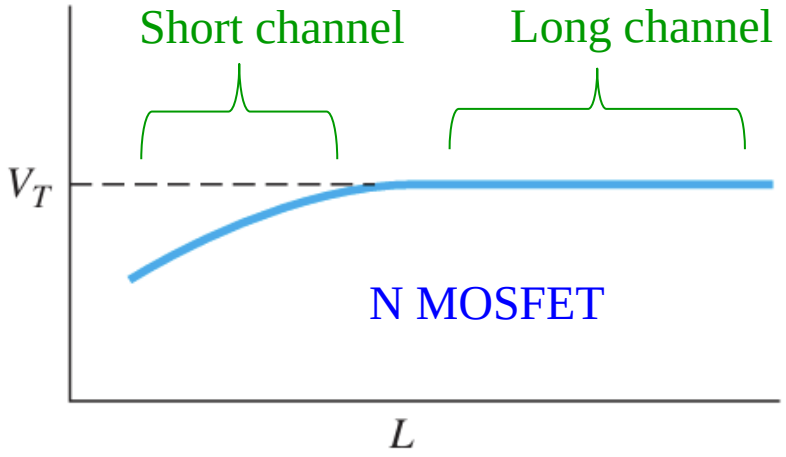
For long or Width channel MOSFET:

$$V_{TN} = (|Q'_{SDmax}| - Q'_{SS})\frac{t_{ox}}{\epsilon_{ox}} + \phi_{ms} + 2\phi_{fp}$$

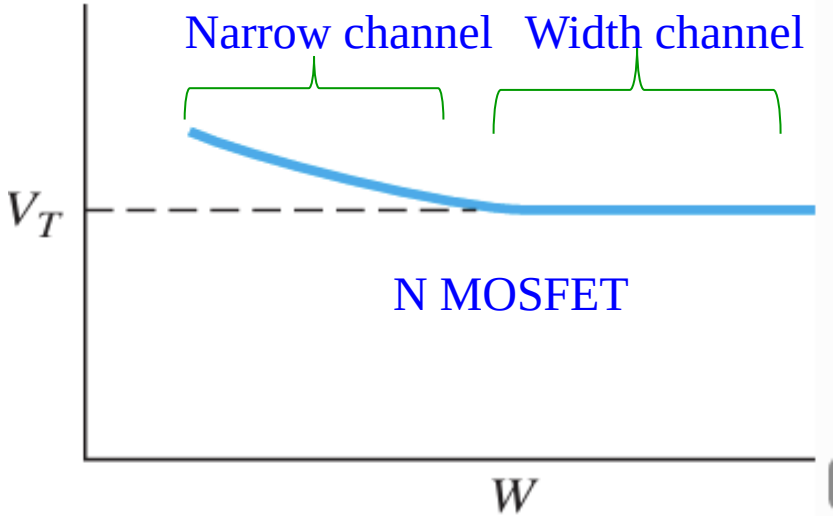
which have no relation with the L and W



V_T vs L for short channel MOSFET（L～r_j）

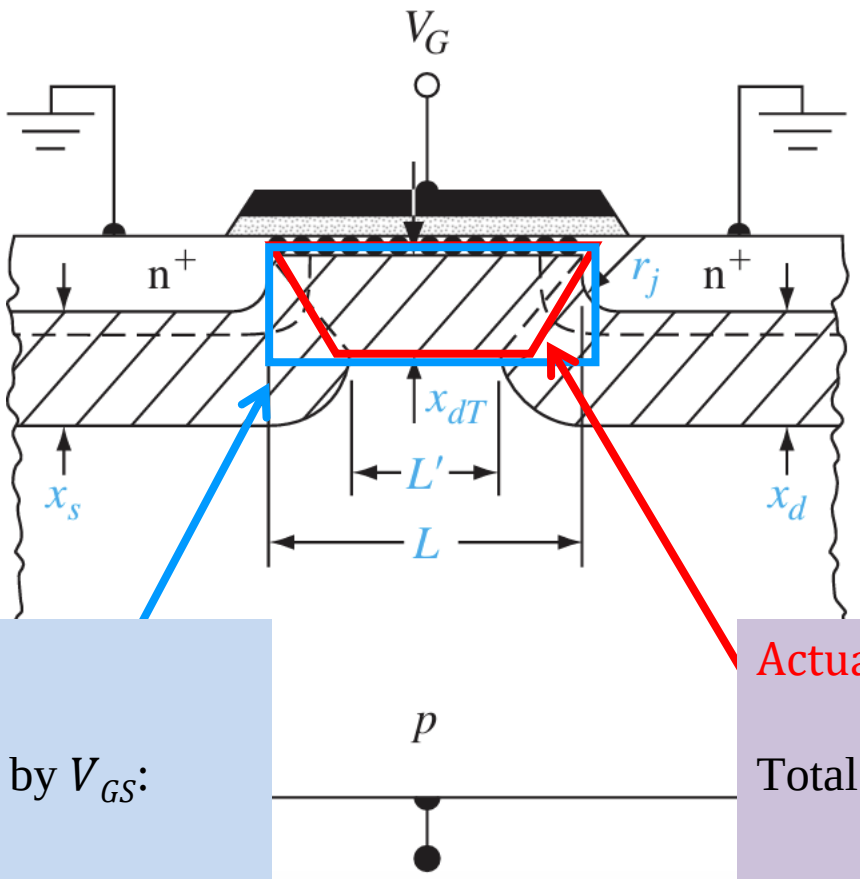


V_T vs W for narrow channel MOSFET（W～x_{dT}）



11.3.1 Short-Channel Effects

V_T varies with L： surface space charge



Ideal model (for long channel):

Total surface charge controlled by V_{GS}:

$$|Q_B| = eN_a x_{dT} WL$$

Surface charge per unit area:

$$|Q_{B\max}| = eN_a x_{dT}$$

Actual model (for short channel):

Total surface charge controlled by V_{GS}:

$$|Q'_B| = eN_a x_{dT} \frac{L + L'}{2} W$$

Surface charge per unit area:

$$|Q'_{B\max}| = eN_a x_{dT}(1 - \frac{\Delta L}{L})$$

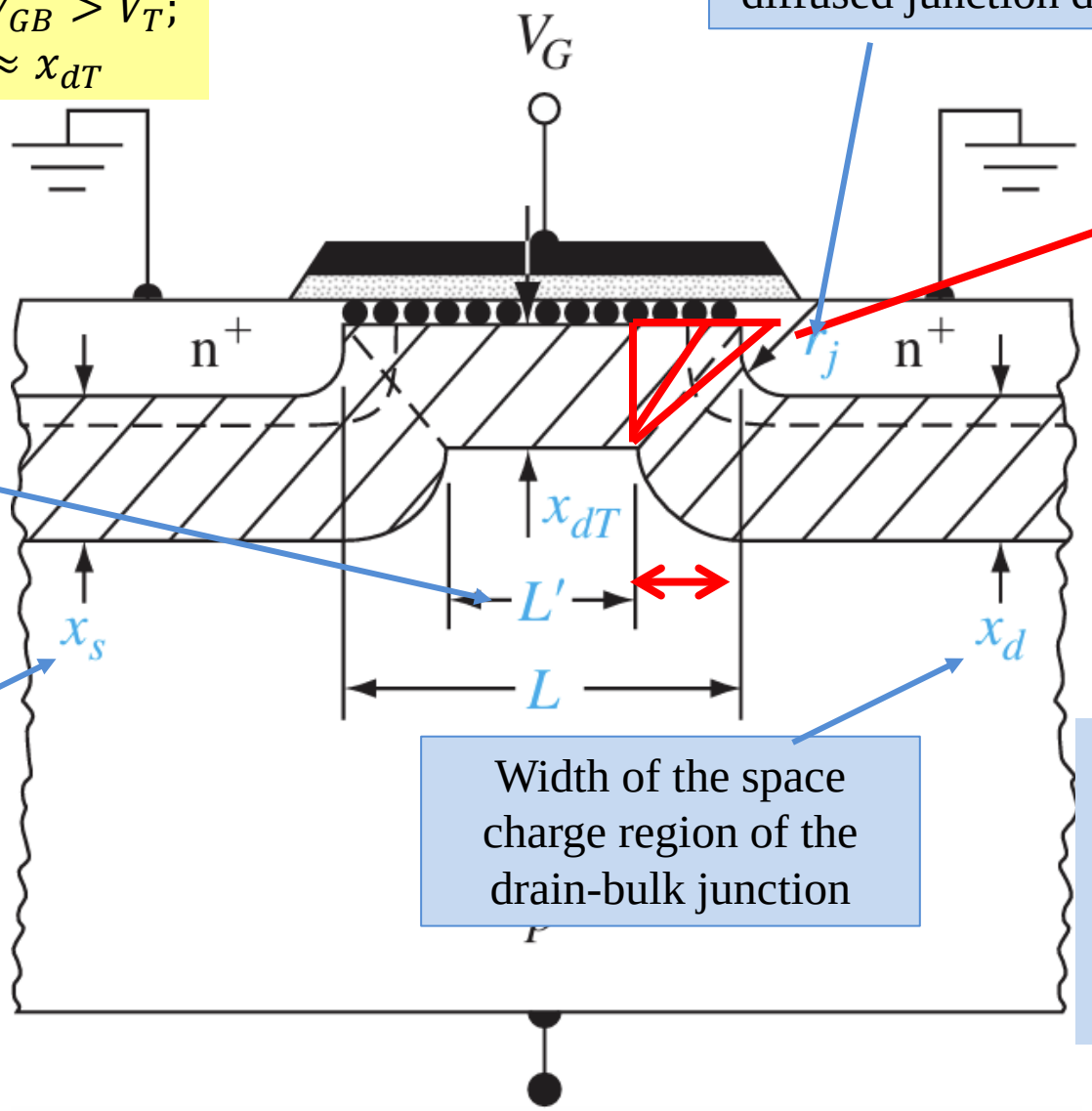
$$(L' = L - 2\Delta L)$$

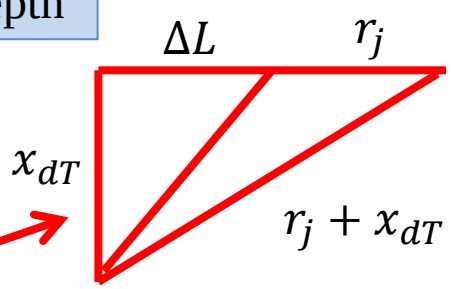
11.3.1 Short-Channel Effects

V_T varies with L： How to solve ΔL ?

Assumption: (1) $V_{DS} = 0, V_{GB} > V_T$;
(2) $x_s \approx x_d \approx x_{dT}$

diffused junction depth





Width of surface space charge region

Width of the space charge region of the source-bulk junction

Width of the space charge region of the drain-bulk junction

$$\Delta L = r_j\left(\sqrt{1+\frac{2x_{dT}}{r_j}}-1\right)$$

$$|Q'_{B\max}| = eN_a x_{dT}\left(1-\frac{\Delta L}{L}\right)$$

$$= eN_a x_{dT}\left[1-\frac{r_j}{L}\left(\sqrt{1+\frac{2x_{dT}}{r_j}}-1\right)\right]$$

11.3.1 Short-Channel Effects

V_T varies with L： How to solve ΔL ?

$$V_T = (|Q'_{SDmax}| - Q'_{SS})\frac{t_{ox}}{\epsilon_{ox}} + \emptyset_{ms} + 2\emptyset_{fp} = \emptyset_{ms} + 2\emptyset_{fp} + \frac{eN_a x_{dT}}{C_{ox}}\left[1 - \frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right]$$

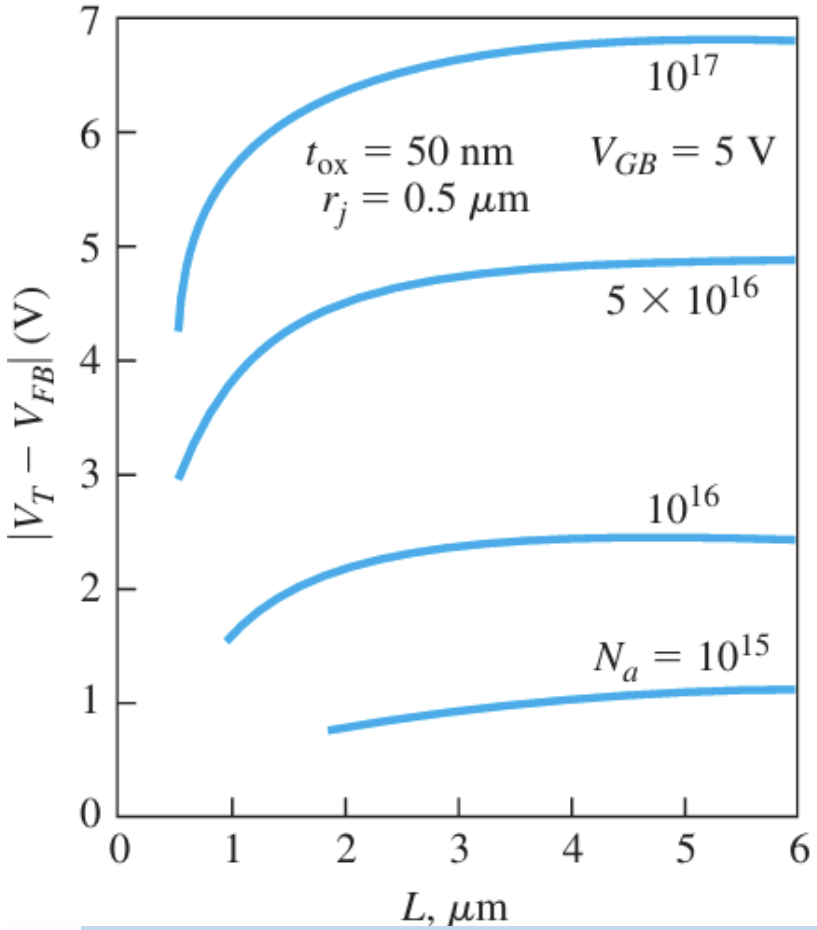
$$\Delta V_T = V_T\text{（}short\ channel\text{）} - V_T\text{（}long\ channel\text{）} = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{r_j}{L}\left(\sqrt{1 + \frac{2x_{dT}}{r_j}} - 1\right)\right] < 0$$

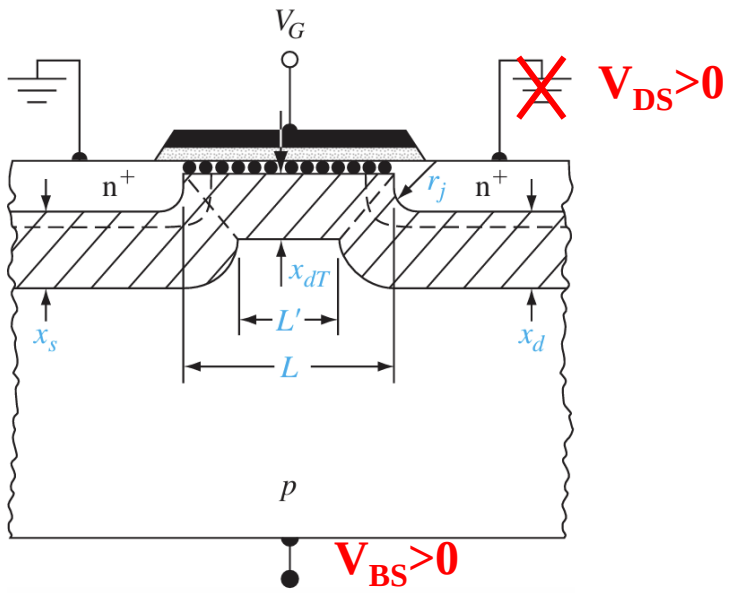
When L is comparable to the junction depth r_j , V_T decreases with decreasing L;

Calculate the threshold voltage shift due to short−channel effects:
n MOSFET, $N_a = 3 \times 10^{16}\text{cm}^{-3}, t_{ox} = 45\text{nm}, L = 1.25\mu\text{m}, r_j = 0.5\mu\text{m}$
$\Rightarrow C_{ox} = 7.67 \times 10^{-8}\text{F/cm}^3, \varphi_{fp} = 0.376\text{V}, x_{dT} = 0.18\mu\text{m} \Rightarrow \Delta V_T = -0.140\text{V}$

11.3.1 Short-Channel Effects

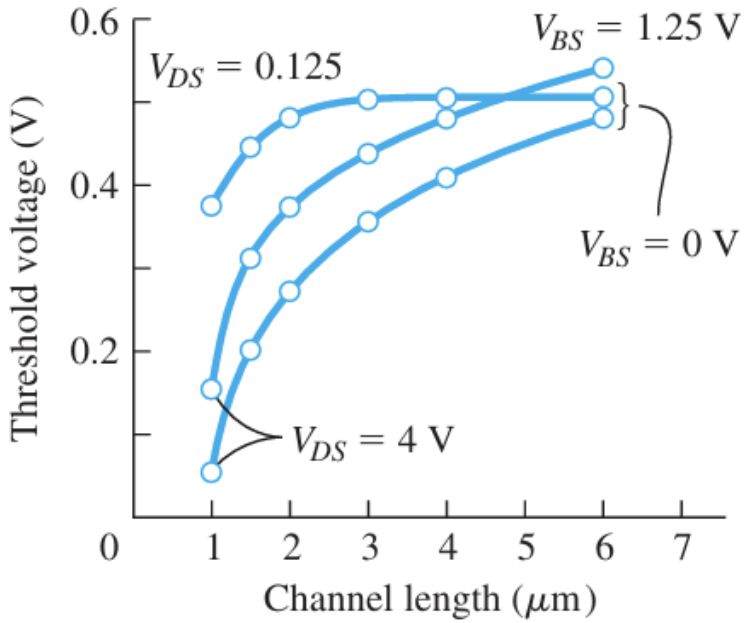
V_T - L curve





$V_{DS} \uparrow \Rightarrow \Delta V_T \downarrow$

$V_{BS} \uparrow \Rightarrow \Delta V_T \uparrow$

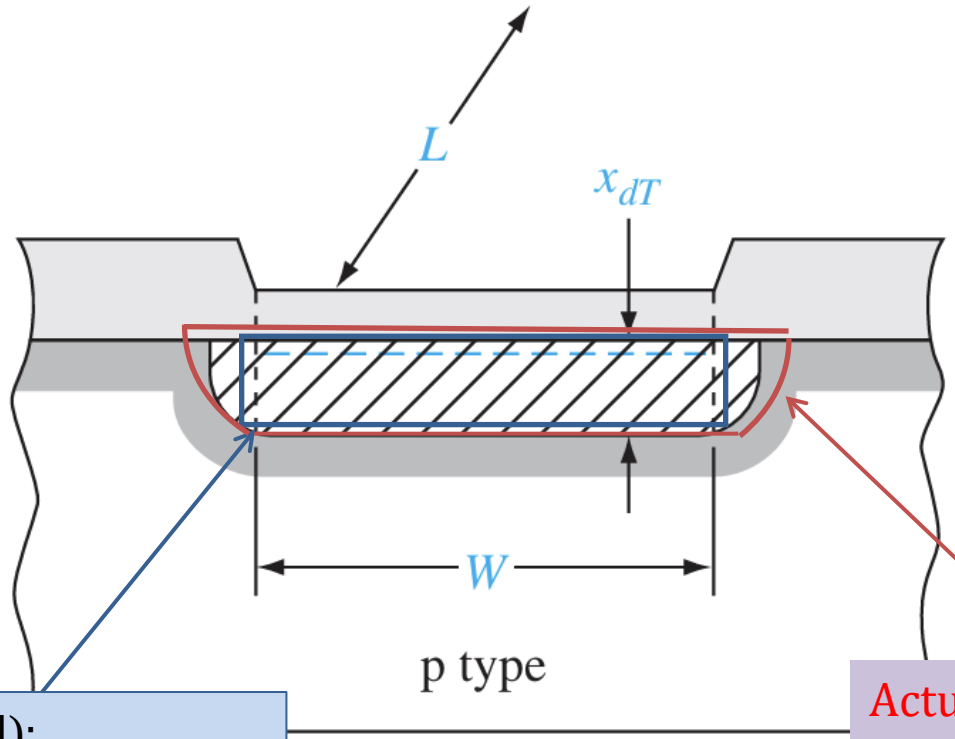


$$\frac{\text{Channel length}}{\text{S} - \text{D diffusion junction depth}} = \frac{\text{L}}{\text{r}_\text{j}} \downarrow \Rightarrow |\Delta V_T| \uparrow;$$

base doping concentration $N_a \uparrow \Rightarrow$ Surface charge density $\uparrow \Rightarrow |\Delta V_T| \uparrow$

11.3.2 Narrow-Channel Effects

V_T varies with W： surface space charge



The volume of the two $\frac{1}{4}$ cylinders at the edge is: $2 \cdot \frac{1}{4}\pi x_{dT}^2 \cdot L$

Ideal model (for width channel):

Total surface charge controlled by V_{GS}:

$$|Q_B| = eN_a x_{dT} WL$$

Surface charge per unit area:

$$|Q_{B\max}| = eN_a x_{dT}$$

Actual model (for narrow channel):

Total surface charge controlled by V_{GS}:

$$|Q'_B| = eN_a x_{dT}(WL + \frac{\pi}{2} L x_{dT})$$

Surface charge per unit area:

$$|Q'_{B\max}| = eN_a x_{dT}(1 + \frac{\pi}{2}\frac{x_{dT}}{W})$$

11.3.2 Narrow-Channel Effects

V_T varies with L： How to solve ΔT ?

$$V_T = \emptyset_{ms} + 2\emptyset_{fp} + \frac{eN_a x_{dT}}{C_{ox}}\left[1 + \frac{\pi}{2}\frac{x_{dT}}{W}\right]$$

$$\Delta V_T = V_T(narrow\ channel) - V_T(width\ channel) = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{\pi}{2}\frac{x_{dT}}{W}\right] > 0$$

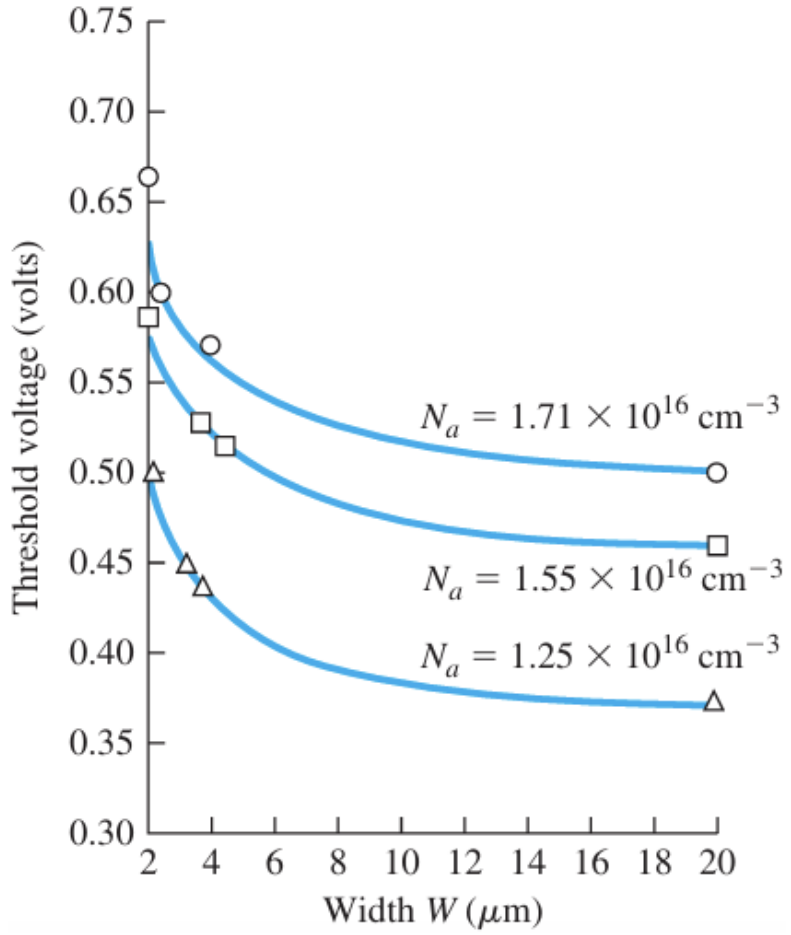
When the channel edge is not 1/4 cylinder: $\Delta V_T = -\frac{eN_a x_{dT}}{C_{ox}}\left[\frac{\xi x_{dT}}{W}\right] > 0$

When W is comparable to the junction depth r_j , V_T increases with decreasing W;

Calculate the threshold voltage shift due to narrow−channel effects:
n MOSFET, $N_a = 3 \times 10^{16}\text{cm}^{-3}, t_{ox} = 45\text{nm}, W = 1.59\mu\text{m}, r_j = 0.5\mu\text{m}, \xi = \pi/2$
$\Rightarrow C_{ox} = 7.67 \times 10^{-8}\text{F/cm}^3, \varphi_{fp} = 0.376\text{V}, x_{dT} = 0.18\mu\text{m} \Rightarrow \Delta V_T = 0.2\text{V}$

11.3.2 Narrow-Channel Effects

V_T - L curve

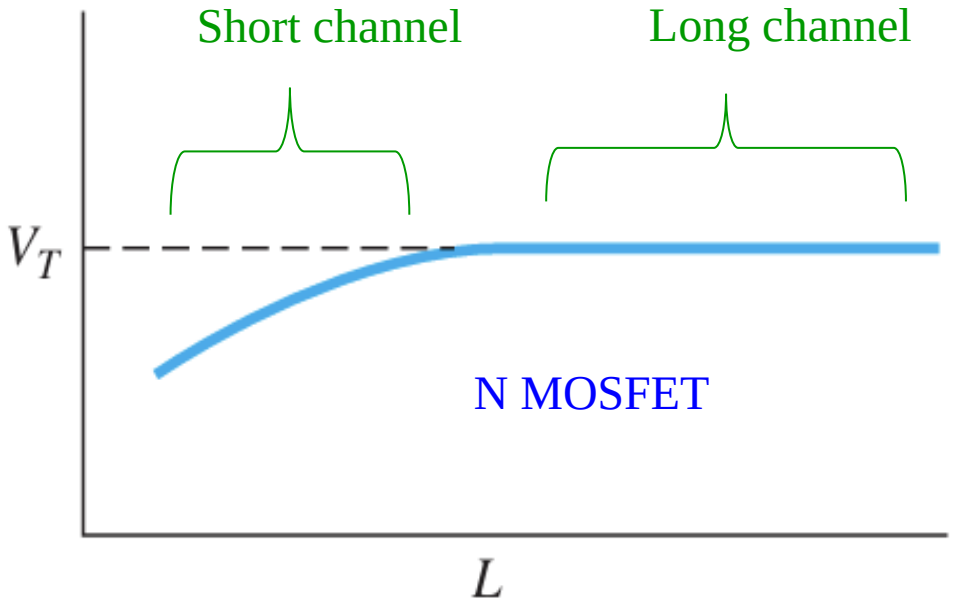


$$\frac{\text{Channel width}}{\text{space charge the vertical width}} = \frac{L}{x_{dT}} \downarrow \Rightarrow |\Delta V_T| \uparrow;$$

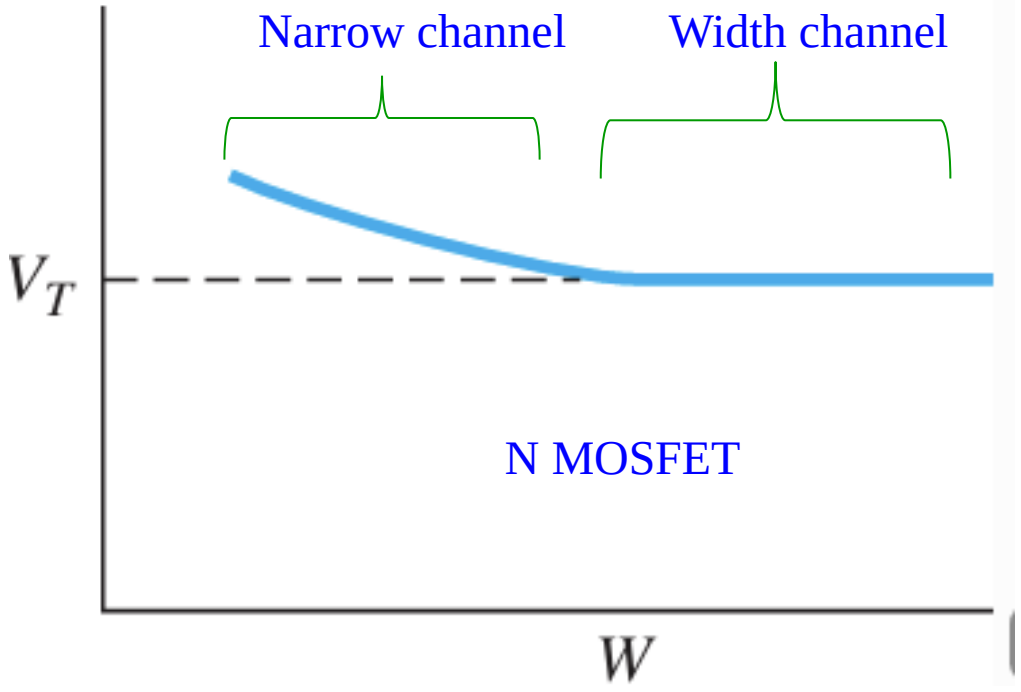
base doping concentration $N_a \uparrow \Rightarrow$ Surface charge density $\uparrow \Rightarrow |\Delta V_T| \uparrow$

11.3 Summary: V_T varies with L or W

V_T vs L for short channel MOSFET（$L \sim r_j$）



V_T vs W for narrow channel MOSFET（$W \sim x_{dT}$）

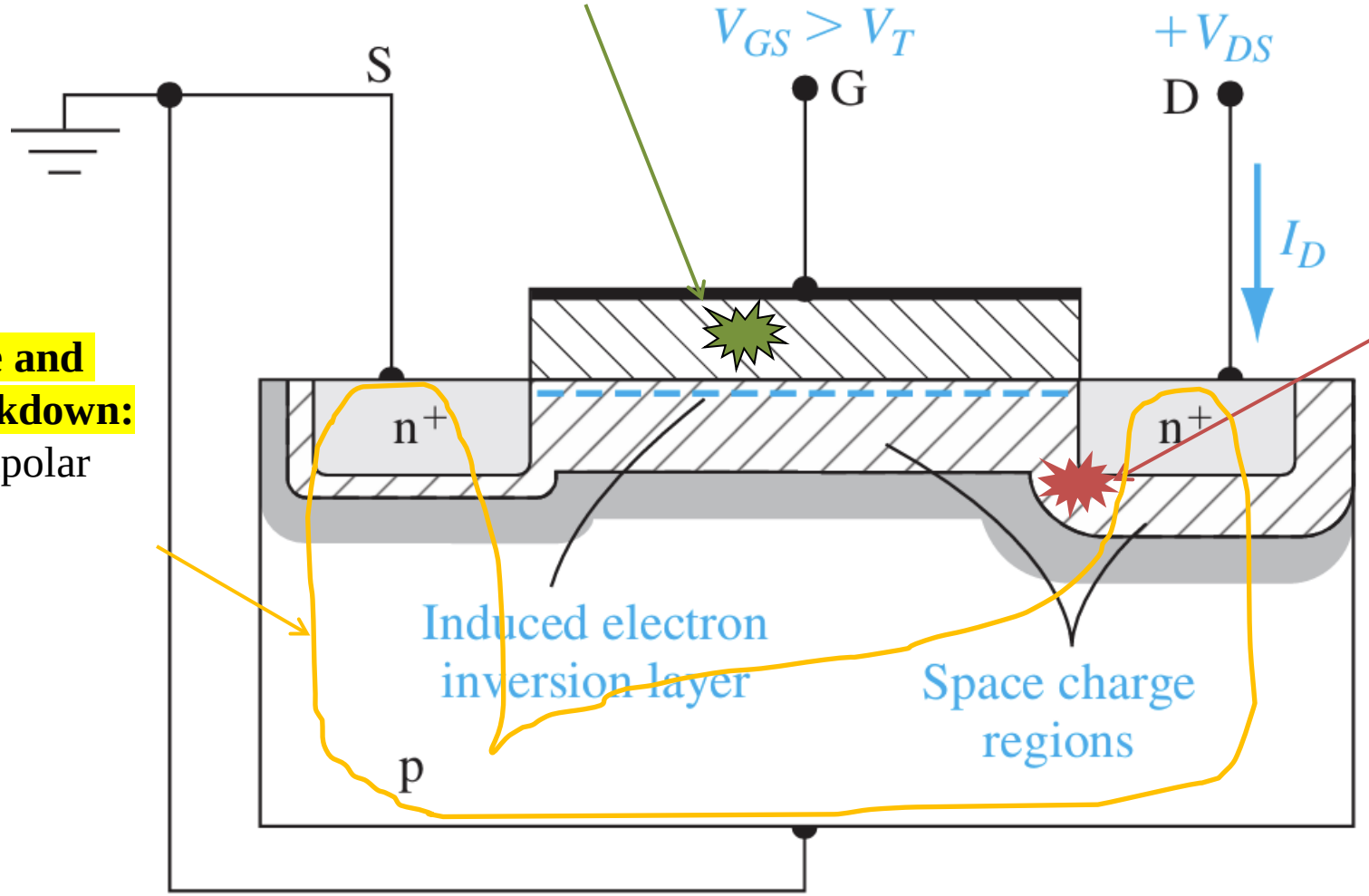


11.4 ADDITIONAL ELECTRICAL CHARACTERISTICS

11.4.1 Breakdown Voltage

The main breakdown

Oxide Breakdown: is only related to V_{GS}



Near Avalanche and Snapback Breakdown: parasitic NPN bipolar transistor

Avalanche Breakdown: The drain pn junction is related to V_{DS} and V_{GS}

Near Punch-Through Effects: Drain-Induced Barrier Lowering (DIBL).

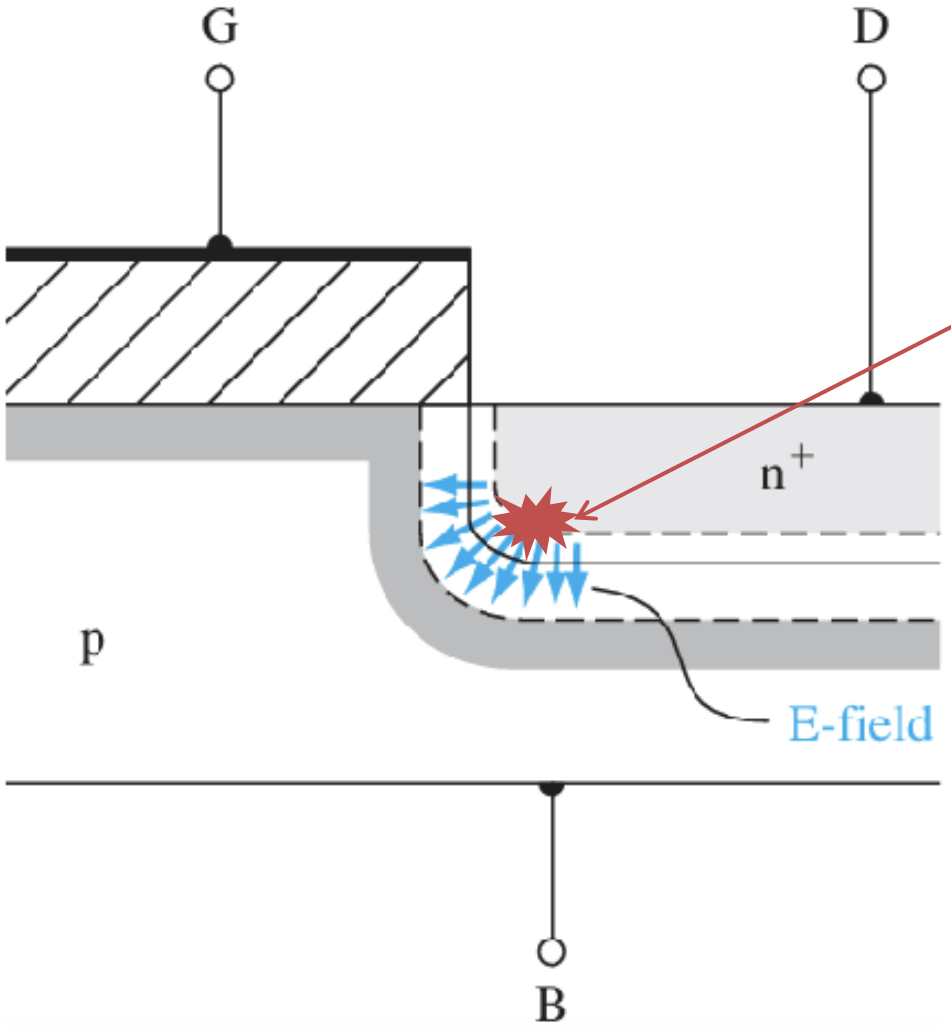
11.4.1 Breakdown Voltage

Oxide Breakdown

- Breakdown phenomenon
  - When $V_{GS}\uparrow = BV_{GS}$ → the electric field of the oxide layer Eox > the critical electric field $E_B = (0.5\sim1) \times 10^7$V/cm, the oxide layer breakdown
  - When the oxide layer thickness is 50nm, BV_{GS}=30V, if $E_B = 6 \times 10^6$V/cm, the working voltage $V_{GS} < 10$V (safety margin is 3)
- Breakdown process
  - Pinholes→ pits→ cavities → collapse
  - Current I↑→ temperature T↑→ current I↑, form a thermoelectric positive feedback
- The source of the breakdown electric field
  - V_{GS}: $E_{OX} \approx V_{GS}/T_{OX}$
  - Gate induces charge: $E_{OX} \approx Q_I/T_{OX}C_{OX}$

11.4.1 Breakdown Voltage

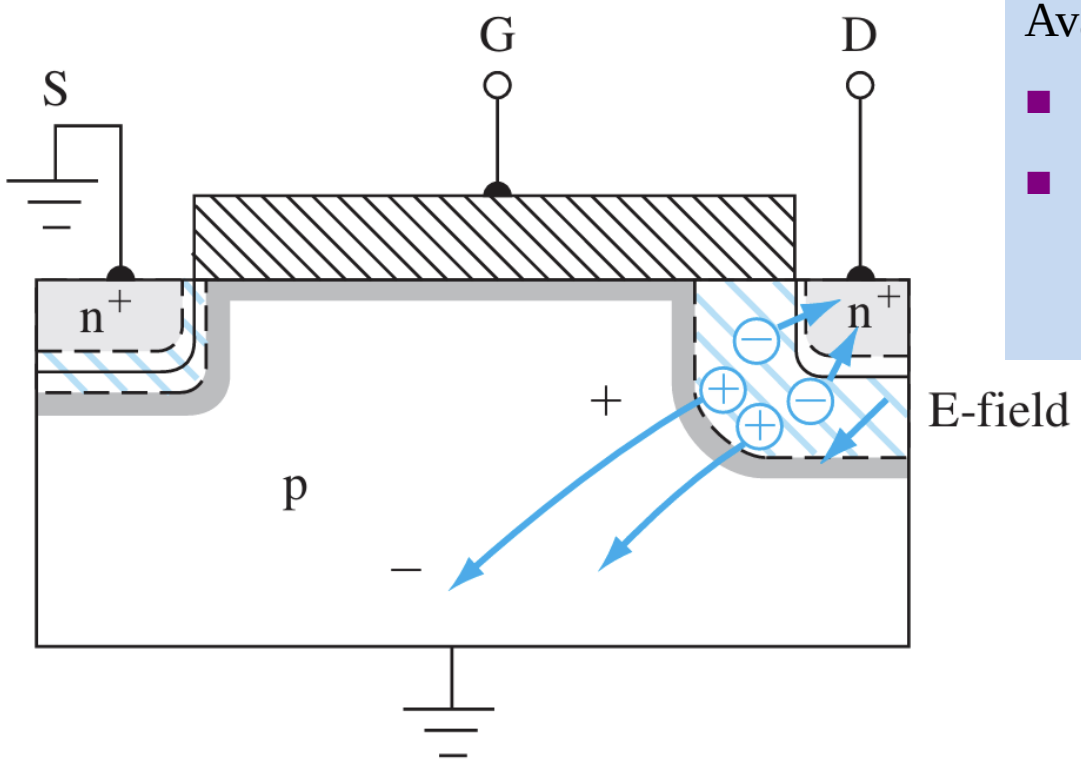
Avalanche Breakdown: drain to base pn junction breakdown



The n^+ drain contact may be a fairly shallow diffused region with a large curvature. The electric field in the depletion region tends to be concentrated at the curvature, which lowers the breakdown voltage.

11.4.1 Breakdown Voltage

Avalanche Breakdown: Avalanche multiplication effect in the channel



Avalanche multiplication conditions:

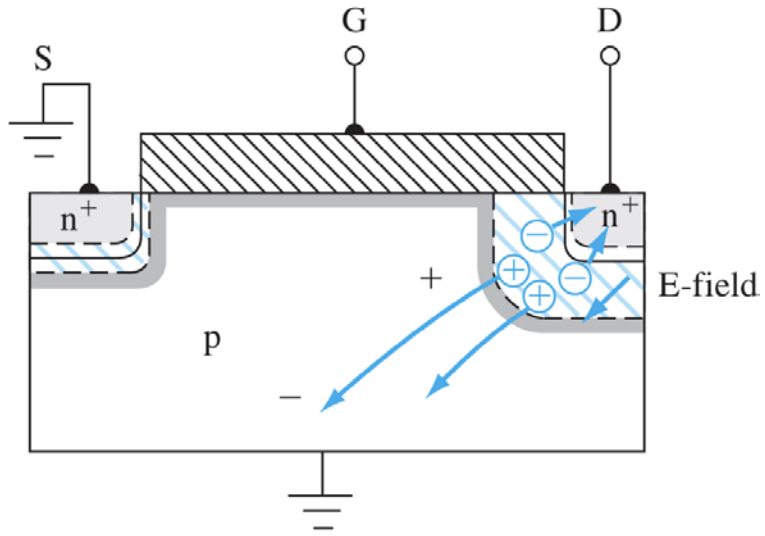
- Short channel: Shorter L, Stronger E in channel
- N channel: compared with electron, the collisional ionization rate of holes is smaller, and the critical electric field that produces avalanche multiplication is stronger.

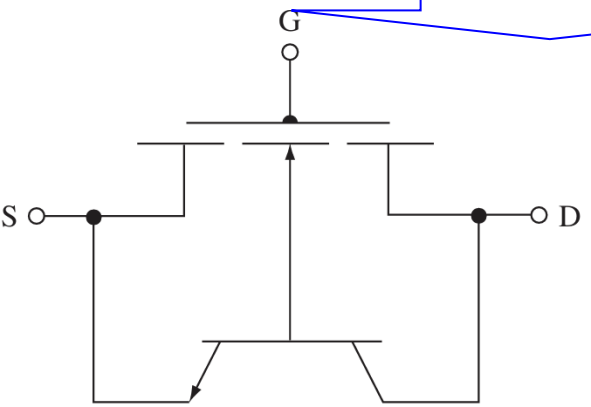
$I_D = M_1 I_S$
(multiplication factor $M_1 >> 1$)

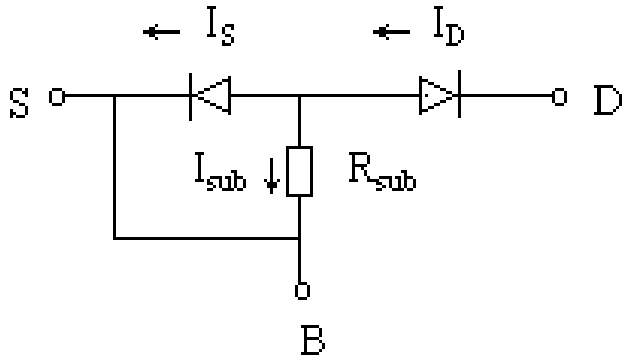
The carrier from the S terminal (forming the current I_S) is accelerated by the channel electric field and an avalanche multiplies near the D terminus, and the resulting electrons are collected by the drain (adding I_D) and the resulting holes are injected into the substrate (generating I_{sub})

11.4.1 Breakdown Voltage

Near Punch-Through Effects : parasitic NPN bipolar transistor







$$I_{Sub} = I_D - I_S = (M_1 - 1)I_S \approx M_1 I_S \quad (M_1 >> 1)$$

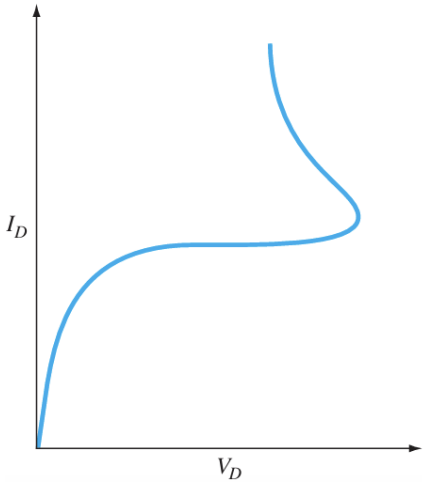
$I_D \uparrow \Rightarrow I_{sub} \uparrow \Rightarrow V_{sub} = I_{sub} R_{sub} \uparrow$

$\Rightarrow$ electrons injected into the channel from the $S \uparrow \Rightarrow I_D \uparrow$

The VDS ↓ required to maintain a high current ID

$\Rightarrow$ a negative resistance effect

(This $I_D \uparrow$ is not caused by $V_{DS} \uparrow$)



Formation conditions:

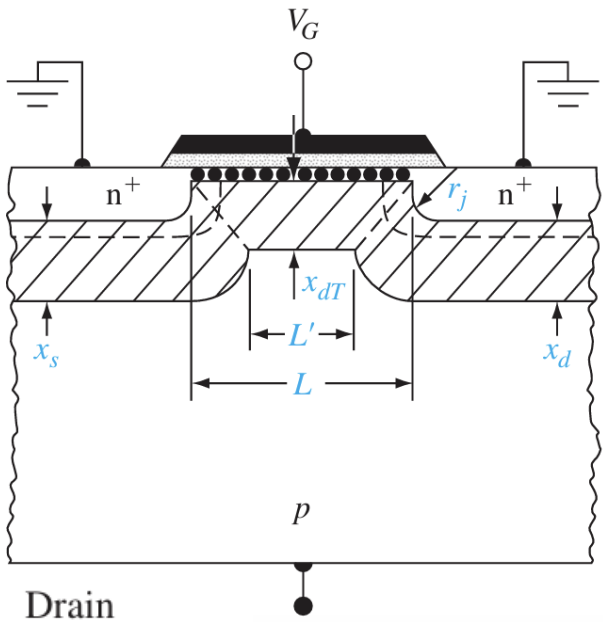
(1) $R_{sub} \neq 0$;

(2) channel avalanche multiplication

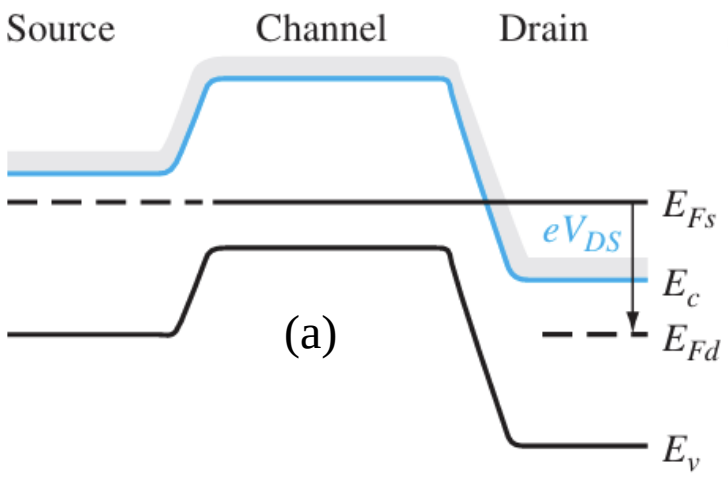
(Short Channel, N channel)

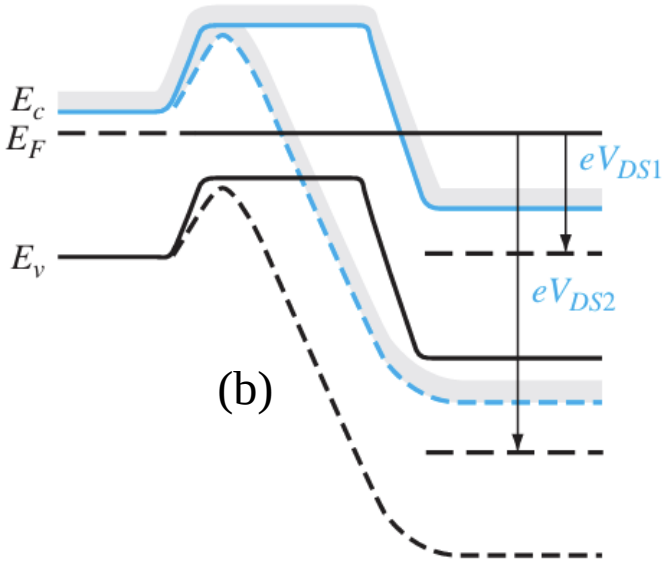
11.4.1 Breakdown Voltage

Near Avalanche and Snapback Breakdown: parasitic NPN bipolar transistor



the DB space charge region extends completely across the channel region to the SB space charge region.
in this situation, the barrier between the S and D is completely eliminated and a very large I_D would exist.



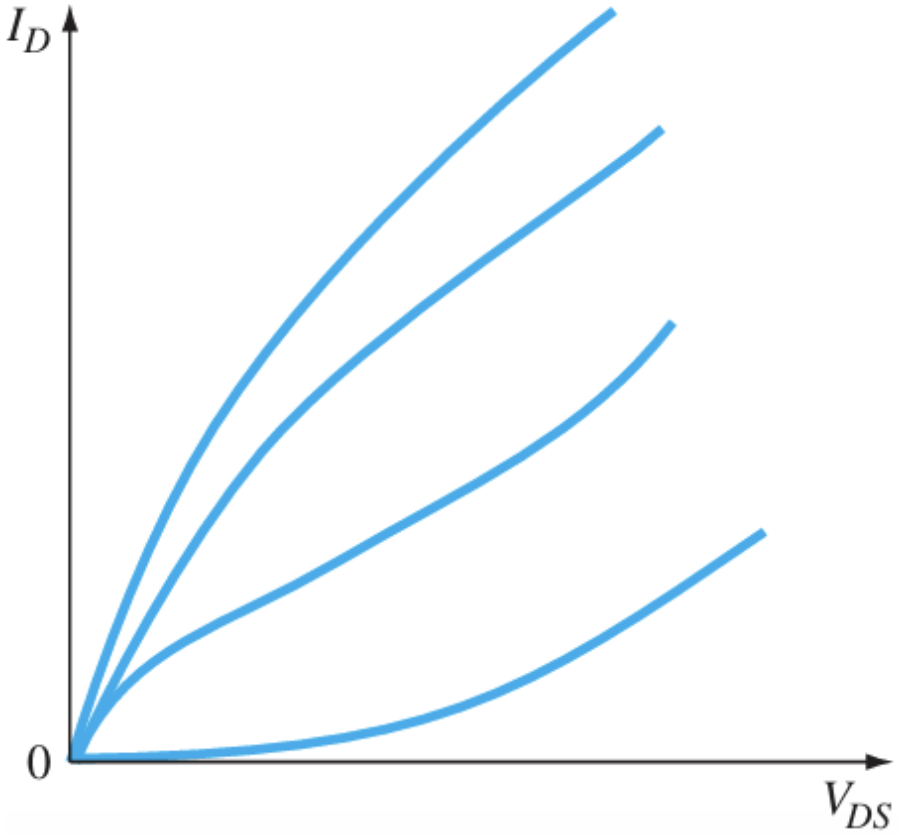


(a) Equipotential plot along the surface of a **long-channel** MOSFET.
(b) Equipotential plot along the surface of a **short-channel** MOSFET before and after punch-through

11.4.1 Breakdown Voltage

Near Avalanche and Snapback Breakdown

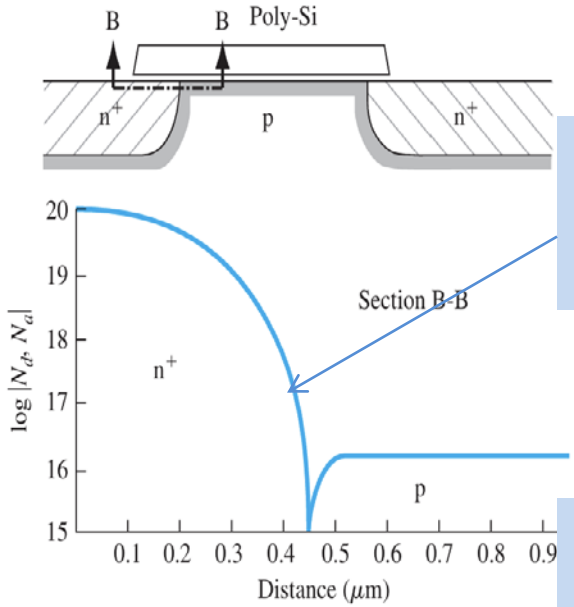
Short-channel device with a near punch-through condition



Typical I–V character istics of a MOSFET exhibiting punch through effects

11.4.2 The Lightly Doped Drain(LDD) Transistor* (aim: reduces these breakdown effects)

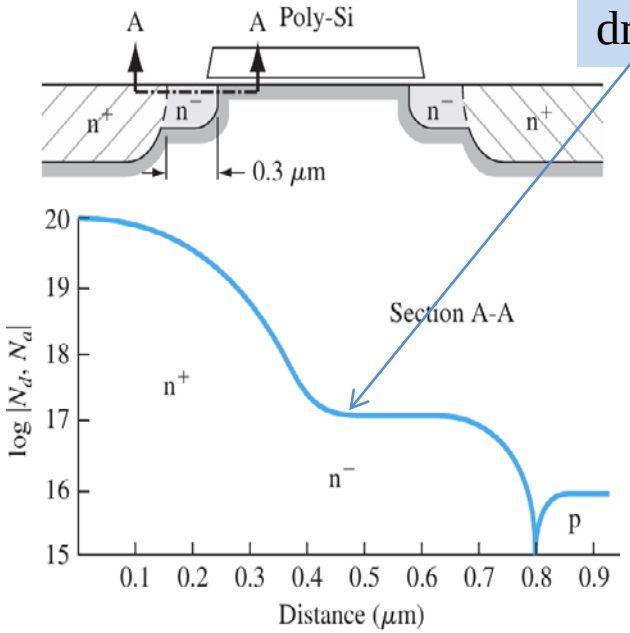
Conventional structure

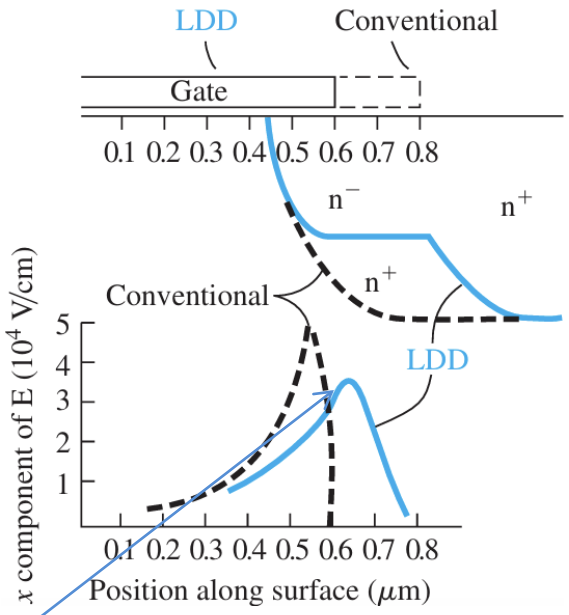


higher doping concentration and steeper distribution gradients in drain

lower doping concentration and shallower distribution gradient in drain

LDD structure





Reduced electric field peaks and distribution gradients

LDD disadvantages: an increase in both fabrication complexity and drain resistance.

11.4.3 Threshold Adjustment by Ion Implantation

- Acceptor impurities N_a (B) are implanted into the surface of the P-sub semiconductor → final doping concentration on the semiconductor surface ↑ → More difficult to form an inverted layer→ V_T↑

Before Implant

Implanted acceptor impurity dose (number of ions injected per unit area）

$$V_T = V_{T0} + \frac{eD_I}{C_{ox}}$$

- Donor impurities N_d (P) are implanted into the surface of the P-sub semiconductor → final doping concentration on the semiconductor surface ↓→ More difficult to form an inverted layer→ V_T ↓

Before Implant

Implanted donor impurity dose (number of ions injected per unit area）

$$V_T = V_{T0} - \frac{eD_I}{C_{ox}}$$

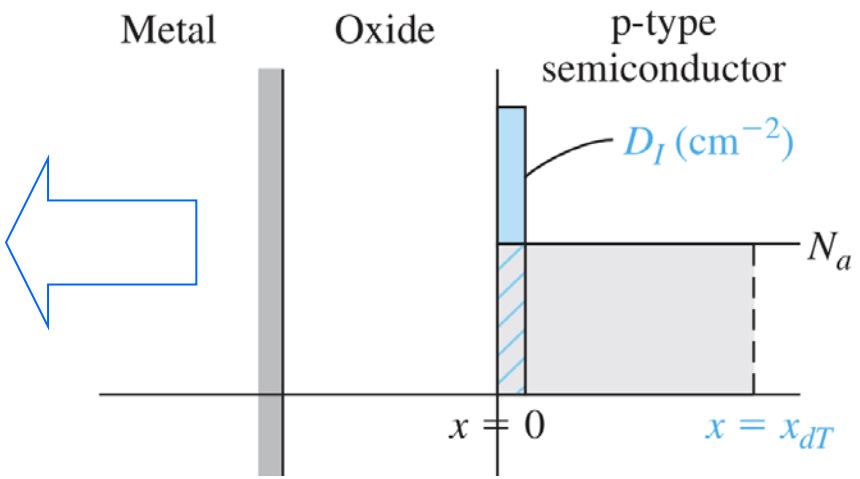
11.4.3 Threshold Adjustment by Ion Implantation

Implant impurity distribution

1. Delta function distribution

$$D_I = \delta(x)(N_S - N_a)$$

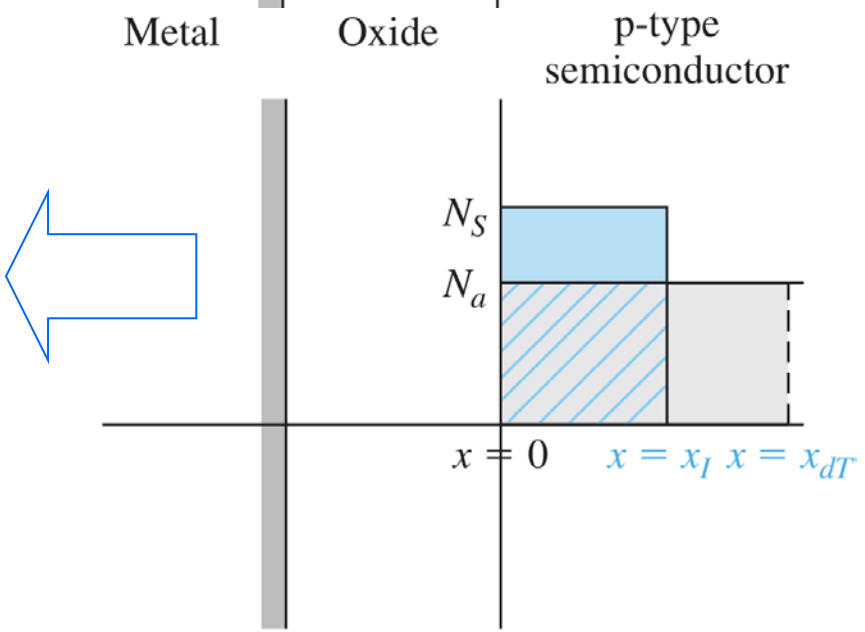
- Average implantation doping concentration
- before implant



2. Step function distribution

$$D_I = (N_s - N_a)x_I$$

- Average implantation doping concentration
- Implantation depth
- before implant



3. Gaussian function distribution：

It is closer to the actual situation, and the analysis is more complex.

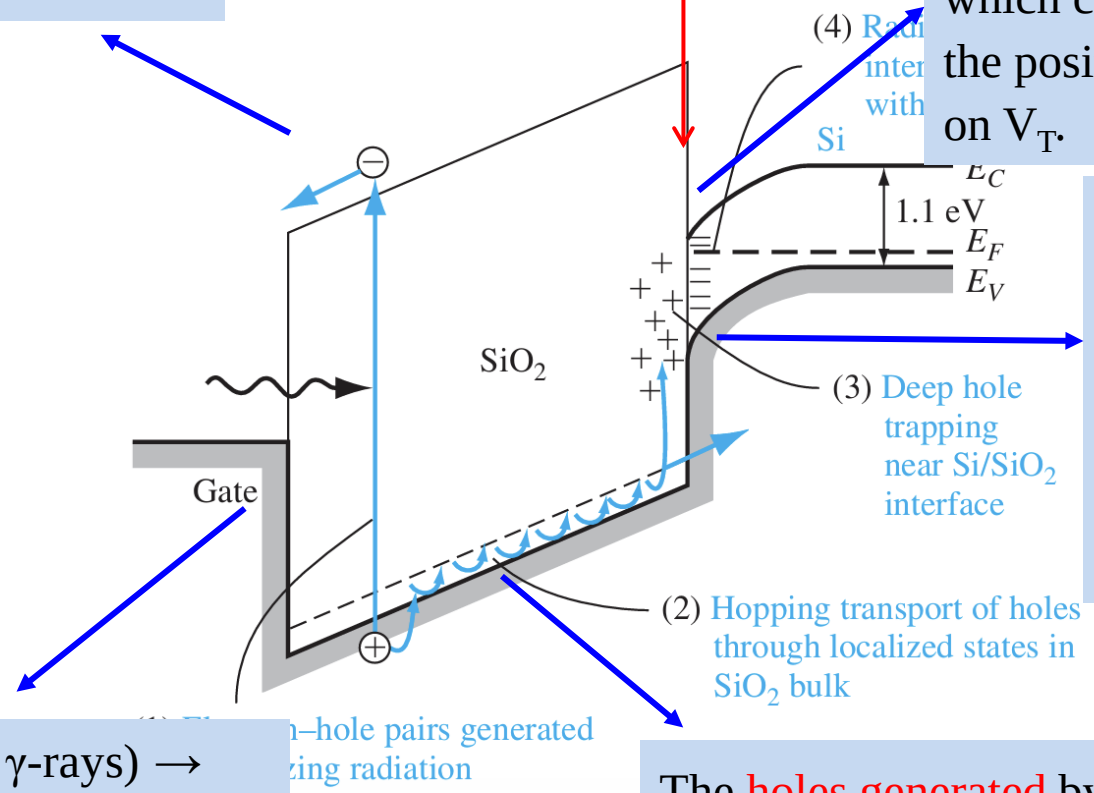
When invented， $X_{dT} < X_I$, V_T is determined by N_s;
When invented， $X_{dT} > X_I$, V_T is determined by D_I;

11.5 * | RADIATION AND HOT-ELECTRON EFFECTS

The electrons generated by the radiation swiftly migrate towards the gate electrode, traversing through the SiO_2 layer. (mobility～20cm²/V•s）

Radiation-Induced Interface States

The ionizing radiation breaks the saturated bonds at the Si-SiO_2 interface → interface traps. These traps manifest as donor states located below the forbidden band and acceptor states above it, which can partially compensate for the effects of the positive oxide charge introduced by radiation on V_T.



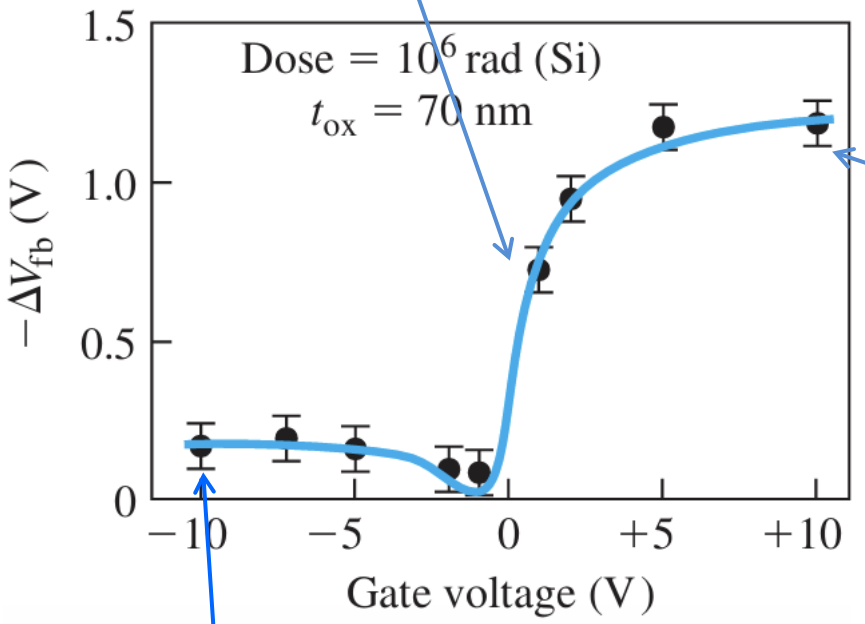
Part of the holes that reach the Si-SiO interface are injected into the Si, while the remaining part becomes trapped by the hole traps located near the interface. This trapping → a positive space charge → V_T to shift in a negative direction.

Ionizing radiation(X-rays and γ-rays) → electron-hole pairs in SiO_2→ free electrons and holes

The holes generated by radiation slowly migrate towards the Si-SiO_2 interface via random transitions within the SiO_2 layer（mobility 10^{-4}～10^{-11}cm²/V•s）

11.5.1 * Radiation-induced oxide charge: Characteristic 1

$V_G > 0$, the holes introduced by radiation move towards base, when $V_G \uparrow \rightarrow$ the holes that reach the vicinity of the Si-SiO_2 interface without being recombined along the way and are eventually trapped by the traps $\uparrow \rightarrow$ additional positive charge $\uparrow \rightarrow$ the flatband voltage shift $|\Delta V_{fb}| \uparrow$.

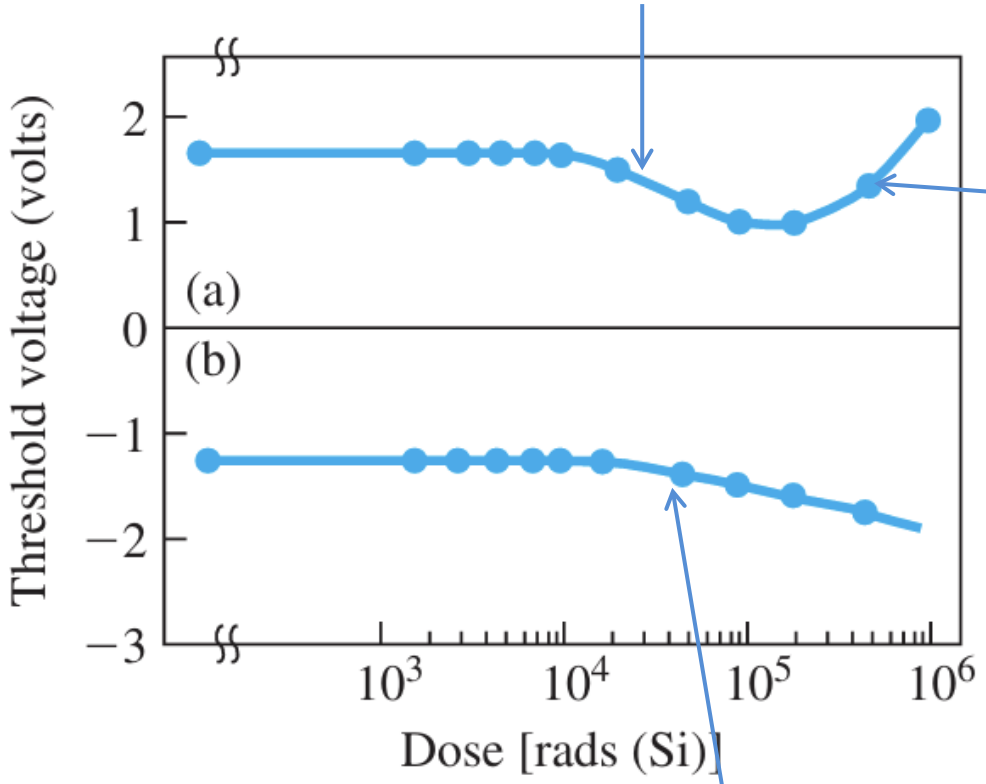


When all the hole traps near the Si-SiO_2 interface are fully occupied, the flatband voltage shift $|\Delta V_{fb}|$ tends to saturate.

$V_G < 0$, the holes introduced by radiation move towards gate $\rightarrow$ a weaker effect of introducing additional positive charge, and this effect remains largely unchanged as V_G varies.

11.5.1 * Radiation-induced oxide charge: Characteristic 2

NMOSFET: positive ON voltage, the effect of radiation-induced oxide charge is strong.

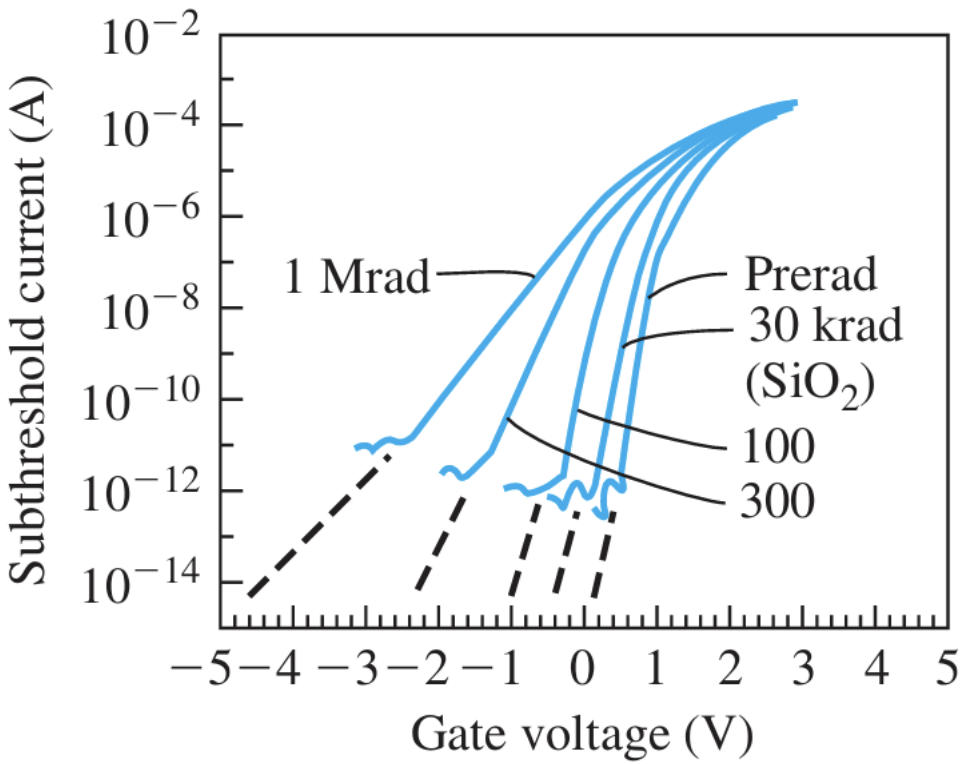


High radiation doses, the radiation-introduced interface states are generated, resulting in a reversal of the V_T shift.

PMOSFET: negative ON voltage, the effect of radiation-induced oxide charge is weak.

11.5.2 * Radiation-Induced Interface States: Characteristic 1

In the subthreshold region, the slope of the I_D-V_{GS} curve is a function of the interface state density.

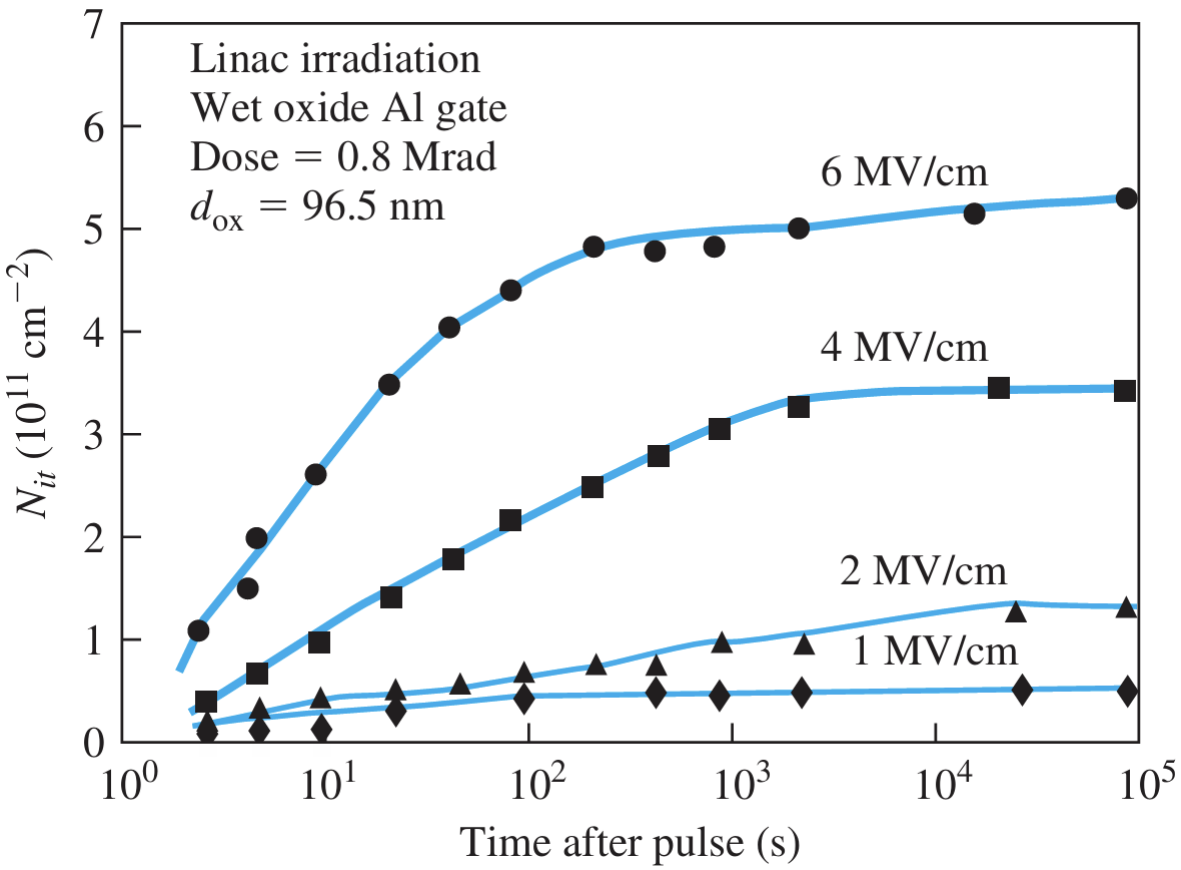


Subthreshold current versus gate voltage of a MOSFET prior to irradiation and at four total radiation dose levels.

The greater the total radiation dose, the greater the introduced interface state density

11.5.2 * Radiation-Induced Interface States: Characteristic 2

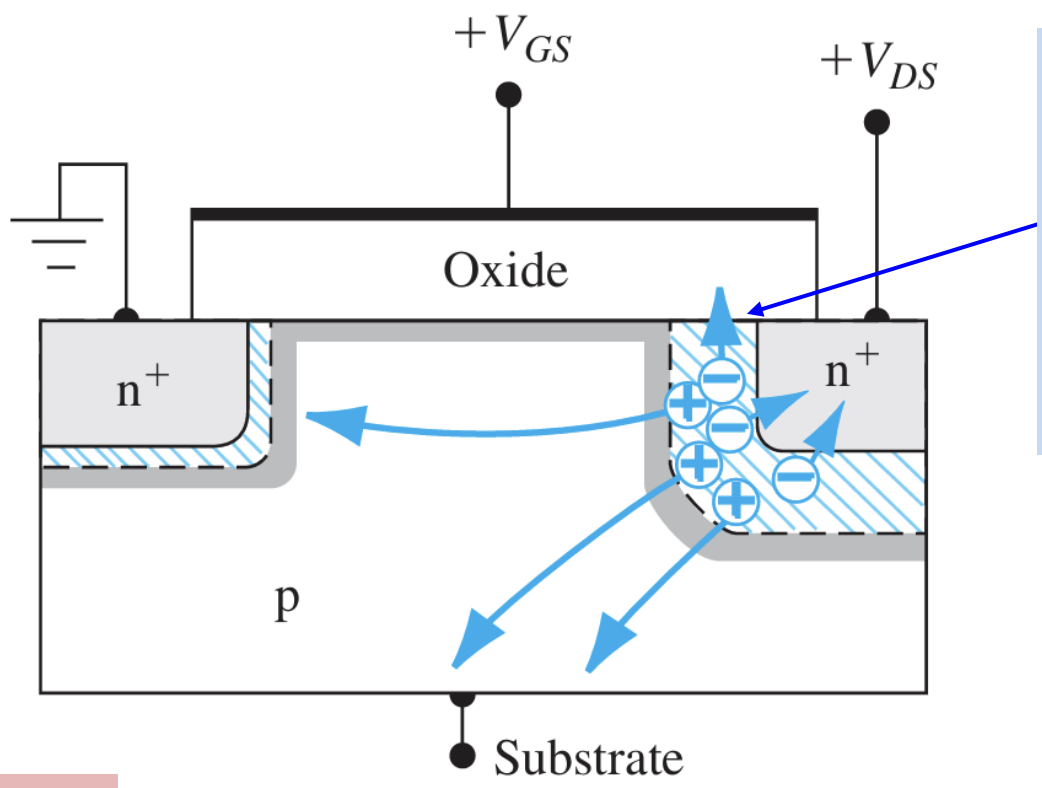
The generation of interface states can also be affected by the electric field in the oxide layer.



Radiation-induced interface state density versus time after a pulse of ionizing radiation for several values of oxide electric field.

After ionizing irradiation, the interface state density gradually increases, and reaches stability after 100 to 1000 seconds.

11.5.3 * Hot-Electron Charging Effects



Hot electrons: High-energy electrons generated by avalanche multiplication in the channel near the drain region may be affected by the longitudinal electric field generated by the positive V_G, cross the Si-SiO_2 interface barrier, and enter the SiO_2 layer. The energy of these electrons is much higher than that at thermal equilibrium.

Function:

- Generate gate current (in the order of pA to fA).
- Produce a negative charging effect, introducing negative oxide layer charge, causing VT to shift positively.
- High-energy hot electrons can generate additional interface states, leading to a decrease in mobility and transconductance.

11.6 Summary

- **Weak inversion region**, $I_{D(sub)}$ exhibits an exponential relationship with V_{GS} and V_{DS}. (**$I_{D(sub)} \propto e^{V_{GS}}$ and $e^{V_{DS}}$**)
- **Saturation region**, channel length modulation occurs, causing a slight increase in I_{DS} with increasing V_{DS}, notably in short-channel or low-doped devices. ($L \rightarrow L' \Rightarrow V_{DS}\uparrow \rightarrow I_{DS}\uparrow$)
- The transverse and vertical electric fields strengthen ⇒ channel mobility decreases; Under strong transverse electric field ⇒ carrier drift velocity saturates ⇒ leading to a linear I_{DS}-V_{GS} relationship.

  ($E_{\parallel}$ and $E_{\perp}\uparrow \rightarrow \mu\downarrow$; $E_{\parallel}\uparrow \Rightarrow v_{drift} \rightarrow v_{sat} \Rightarrow I_{DS} \propto V_{GS}$)
- The **shrink** MOSFET **dimensions** serves as a pivotal strategy to enhance integration density and operational speed.

  (size ↓ → density and speed ↑)
- A **proportional reduction** in device dimensions and operating voltages is desirable but **difficult** to achieve in practical implementation. (size ↓ → density and speed ↑)
- Short or narrow channel conditions, **V_{th}** varies with **L** and **W**. Ion implantation was used to fine-tune V_{th}.
- **Breakdown**(main): GS breakdown and DB breakdown.
- Short channel devices ⇒ **channel avalanche multiplication** ⇒ parasitic transistor effects or hot-electron effects.
- **Ionizing radiation** ⇒ oxide charges and interface states ⇒ **V_{th}** shift and $\mu\downarrow$. The robust design considerations and mitigation strategies to ensure the reliability and performance of MOSFETs.

Exercises:

11.2	11.4
11.15	11.20
11.30	11.42